MNEME VARIANT COMBAT RULES

For the Cepheus Engine System

VARIANT COMBAT RULES

Mneme version 1 Cepheus Variant Combat Rules.

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What is the " Mneme Variant Rules"?

- It takes what would have been Many rolls and boils it to One roll in which the Value is symmetrically interpreted for both its Effects and **outcomes for BOTH the PC and the other party/adversary** or hazards.

Key Benefits of the Mneme Variant Rules are...

- ...Lower Mental Overhead since The Referee does not manage both the Players and the NPCs, but instead handles both using a simpler and easier mechanic with mnemonics make it easier to remember.
- ...Fewer Rolls -- specifically one roll per turn since the Mental Load of the Referee rolling for NPCs and other events is eliminated.
- ...Enables The Referee to manage more players. While there is still a learning curve, the Player management benefit of the Mneme is long term.

Personal Combat Checklist

Personal combat in the Mneme VARIANT Combat System for Traveller System Reference Document is cyclical.

The Referee determines:

1) CONDITIONS which are broken down into the following steps determining:
   a) The Awareness and Visibility of threats, parties, conditions, terrain.
   b) Range, Concealment, Cover, Sizes, Movement
   c) What Actions and Opportunities are available to all parties?
   d) On whose Initiative adversaries act.
2) RESOLVE ACTIONS
   a) The Referee resolves actions based on the initiative of the PC.
   b) The Referee doesn't roll. The assumptions behind this are as follow:
      i) The PCs and their Adversaries all take action simultaneously.
      ii) Adversaries that are engaged in combat with PCs will do damage to the PCs on an outcome of Costly Failure or Costly Success.
   c) The PCs have
      i) Free Actions
      ii) 3 Minor Actions
      iii) 2 of these Minor Actions can be used to perform Significant Actions.
   d) Determine the Outcome of Actions if they fail, succeed, or succeed at a cost.
3) ROLLING - The PC rolls against the Task Difficulty the Referee Assigns after factoring in the Conditions and Actions.
4) OUTCOMES
   a) The Referee and PC take note of which Conditions are changed when applying the outcome of the rolls.
   b) The Referee consolidates the Changes in Conditions and begins a new combat round by describing the new situation.
   c) The Referee declares new combat rounds until the situation determines that combat has come to an end.

Variant Rules

Main Rules Changes

One Roll per Turn

To speed up play and to keep the focus of the game on the back and forth conversation between the Referee and the Players, there will be only one Difficulty Roll per Turn.

Outcomes of this Principle is:

1) **Referees Don't Roll**. The PC rolls, and the Referee sets the Difficulty.
   a) **Failure can result in PC Harm**, Damage, and Negative Outcomes.
   b) **Higher Stakes per Roll**. since failure can result in a lot of harm.
   c) **Focus on the Action of the Turn**. The higher the stakes the more discussion or storytelling about the action.
2) **Fewer Interruptions per Turn**. The Referee doesn't have to switch to too many PCs, actions, new changing factors and conditions per turn.
3) there are more Damage Rolls and Outcome rolls, are pushed to the Players duties freeing up the Referees bandwidth.
4) The effect of actions of NPCs, and other circumstances the PCs do not control are grouped together and resolved using Superiority Mechanics.

Mnemosyne or Mneme - muse of memory. The goal of the system is easier on the mind to run.

I kick down the door and attack the guards.
Okay. Make your roll to kick down the door.
I roll an 8 with Melee.
Okay. Good. The guards will try to dodge your gunfire. Make your attack roll.
Let's see... 8 plus gun combat, less 1... I get an 11.
Roll for damage.
I roll a 4.
The guard takes 3 damage. Now it's his turn.
I won't dodge any attacks.
Well, lucky you. He misses. I'll now roll for the other guard.
The other guard gets a 12 and hits you.
I have 4 ballistic armor.
Noted.
I'm up next, right?
I roll 6 damage. With your armor, you take 2.
Neat!
Nope. It's Jane, then you, then me.

MNEME RESOLUTION

Pro in Comparison
1. Familiar
2. Simpler to run - Less Factors and Options

Con in Comparison
1. Many Rolls -
   a. Resolving NPCs - but if an npc will perform an action its own rolls.
   b. Players who use reactions are more difficult to track. You need to remember before the end fo the turn to remove Reaction Penalties.
2. More than One Character is handled per Turn due to reactions - even NPCs.
3. Injury is more complicated.
4. Doesnt Scale well - More Forces more time and Effort
5. No Economies of Force. Subject to Action Economy - Little room for tactics
6. Stat and Skills used defensively has no Influence in odds. Only in attacking.
7. No grade of effect, most only matter of ^+ Effect

Actions
1. Free Actions
2. Minor Action
   a. Move 6m
   b. Change Stance
      i. Prone
      ii. Crouching
      iii. Standing
   c. Ready
      i. Weapon
      ii. Ammo
      iii. Reload
   d. Aiming +1 DM, up to 6
3. Significant Action
   a. Attack
   b. Skill Action
4. Wait
5. Extended Action

Modifiers
1. Characteristics
2. Skill
3. Difficulty +6 to -6
4. Load (Med/Hvy) -1 to -2
5. Target Range per weapon
6. Weapon DM
   a. Aim per minor action
   b. Weapon Scope
   c. Targeting Software
7. Stance
   a. Standing
   b. Crouch
      i. ½ move,
      ii. Improve cover by next level.
   c. Prone
      i. Improve cover by next level
      ii. -2 DM to be hit by ranged attacks,
      iii. +2 DM to be hit by personal-range attacks
8. # of Reactions -1/reaction
9. Target Reactions
   a. Parry shield + skill
   b. Dodge
10. Target Moving
11. Target Cover
   a. -1 to -4 DM
12. Environmental effects
   a. Weather - 1 DM
   b. Darkness -1 to -4 DM
   c. Smoke/Fog -1 DM, -2 DM for lasers

Traveler SRD

- Ambush
- Leadership/Tactics
- Roll Initiative for every Character
- Resolve Action for every Character
- Determine Conditions
- Determine Actions
- Determine Reactions per PC/NPC
- Determine Modifiers, Hasten & Reaction Penalty
- Roll+DM vs 8+
- Determine Effect
- Roll Damage - Armor
- Determine what Stat gets wounded
- Recalculate Stats / Stat Effects
- Apply Initiative penalty and remove reaction penalty

Mneme Variant

- Awareness
- Roll Initiative for every PC
- Resolve Actions for every PC
- Determine Conditions
- Determine Actions simultaneous with NPC/Adversary
- Determine Modifiers & Difficulty
- Roll + DM - Diff
- Determine Effect
- Roll Damage - Armor
- Check Wound Penalties & Thresholds
- Determine Outcome

Conditions
1. Awareness (Dis/Ad)
2. Range
3. Visibility 1-4
   a. Concealment
   b. Invisible
   c. Blind
   d. Darkness
4. Cover 1-4
5. Environment & Terrain Effects
   a. Gravity
   b. Movement
   c. Visibility
6. Superiority
   a. No. of Adversaries
   b. Firepower
   c. Time
7. Target Size
8. Target Moving
9. Hazzards
   a. Under Fire

Actions
1. Free Action
   a. Quick Phrase/Shout
   b. Quick Glance
2. Minor Action
   a. Grabbing/ Taking/ Readying
   b. Aim
   c. Quick Glance/Scan
3. Significant Action
   a. Action Options
      i. Reckless Action
      ii. Supporting Action
      iii. Coordinate
      iv. Plan/Wait
   b. Defend
   c. Attack
      i. Attack Options
         1. Rate of Attack, Auto Fire
         2. Melee Attack
         3. Ranged Attack
         4. Moving Attack
         5. Targeted
   d. Skill Action
      i. Dash
      ii. Search
      iii. Stealth
      iv. Use Object
4. Full Action
   a. Run

Modifiers
1. Characteristic
2. Skill
3. Weapon Aim
   a. Sights and Scope
   b. Ballistic Correction
4. Superiority
   a. Firepower
   b. Manpower
5. Load
   a. Medium Load -1 DM
   b. Heavy Load -2 DM
6. Wound Penalties, every 3 damage -1.

Pro in Comparison
1. Single Roll per Turn - Simpler.
2. Scales well with the Superiority Mechanic
3. Using Chunking to group concepts into Conditions and Modifiers.
4. Carry Over the same technique in Non-Combat by any High stakes situation.

Con in Comparison
1. Unfamiliar System
2. More Actions and Options.
3. Time to Learn and Share the knowledge.

Core Task Resolution

Referee Doesn't Roll

- Any Risk or Threat or Challenge to the PC is the PC rolling against a Difficulty set by the Referee.
- Damage is rolled by the PCs who are Affected. Players roll the damage that their PCs incur or inflict .

Dice Roll

Roll 2d6 + Modifiers and Compare against the Difficulty. A Dice Roll equal to or greater than the Difficulty is a **Success**; otherwise, it's a **Failure**.

- Dice Roll is the action of rolling Dice plus Modifier.
- Dice Roll is also a noun which means the Value of the Dice plus Modifiers.

Modifiers to Rolls

1) Advantage or Disadvantage
2) Competence Modifier. Characteristic Modifiers and Skills (or lack of Skills which is -3 skill penalty).
3) Equipment, and Actions.

- Modifiers of the same kind do not stack, choose the more relevant or the most favorable
  - Example - multiple scopes and sights do not stack.
  - Some Modifiers are Temporary. Aim Modifier increases as minor actions are used.
- Anything that will affect the Chances of the Player that is neither Action or Equipment fall under Conditions.
- Players need to keep track of their Actions and Equipment modifiers.
- Adversaries Modifiers are conditions that affect the difficulty of the roll.

Characteristic Modifier (ChM)

This is the Modifier from a Characteristics. This is derived by taking the Characteristic divide by 3, rounding down, and -2.

- $[(Characteristic \div 3) - 2] = Characteristic\ Modif$

Skill Modifier (Skill)

This is the modifier granted by the character's skill.

- Unskilled Characters have a -3 to such roll, if the Referee even allows it.

Competence (Cmp)

This is the relevant Characteristic Modifier and Skill Modifier.

Difficulty

A check's Difficulty is a value defaulting at 8 set by the Referee.

- Some Actions or Tasks are Harder because they have more Scope and Scale.
  - Scope refers to the amount of work and complexity it undertakes.
  - Scale is the amount of stakeholders, benefits, and the quality and amount of Outcome, Product, or Results.
- The Referee can Modify Difficulty based on Conditions.
- Modifiers to Difficulty are also called difficulty.
- Conditions modify difficulty.

Stakes

- If there are no stakes no need to roll. Tasks and Actions may just take up time and attention.
- If there are stakes then a roll has something to Gain, to Lose, or a Loss to manage.

Difficulty Number Table:

This assumes no modifiers.

Difficulty	Odds	Difficulty
3	97.22%	
4	91.67%	Easy
5	83.33%	
6	72.22%	Routine
7	58.33%	
8	41.67%	Challenging.
9	27.78%	
10	16.67%	Difficult
11	8.33%	
12	2.78%	Very Difficult
13	NA	
14	NA	Formidable
15	NA	
16	NA	

Effect

This is the Dice Roll minus the Difficulty. The effect measures the Degree of Success or Failure of a roll. The Effect is used to prevent a lot of die rolls in the game.

To determine the Effect the Referee or Player simply subtracts the Difficulty from Dice Roll rolled and its modifiers.

- Rolling 2d6 + Modifiers minus the Difficulty. See Dice Mechanics Diagram

Importance of Effects

1) Because the Referee doesn't roll, the negative outcomes of rolls, Fail Effects, is how to determine if Adversaries successfully Attacks and make Damage Rolls against the PC.
2) Multiple Attacks by Adversaries and PCs are resolved through Effects.
3) Skill level reduces the Amount of Harm or negative effects of Failure.

Fail Effect

When the Effect is less than the Difficulty, the Check results in a Fail Effect. The degree of the Fail Effect is equal to the Difficulty minus the Effect.

- For example, if the Difficulty is 8 and the Dice Roll is 5, the degree of the Fail Effect is 3.
- The greater the degree of the Fail Effect, the worse it will be for the PC3.
  - See Dice Mechanics Diagram

Ordinary Failure

Ordinary Failure is when a failure doesn't cause the PC to suffer any harm, incur any injury, or be exposed to any sort of attack.

- The only thing lost by the failure is the opportunity that was spent.
  - Synonymous to Ordinary Failure is a minor setback, or harmless failure.

Mitigating Factors

The player may use Mitigating Factors (DMs) to reduce the degree of the Fail Effect. When the degree of the Fail Effect is reduced to zero or less, the Fail Effect becomes an Ordinary Failure.
There are three categories of DMs that are applied in the following order:

- First Priority are Conditions like Concealment, Range Size Move Modifiers are first applied to reduce the Fail Effect.
- ***The Second Priority DM is Cover***. It is invoked to reduce the degree of Fail Effect only after the First Priority DMs have been applied.
- Example.
  - The PC rolls to attack with 2d6+6 against a 13 Difficulty. Of that 13 Difficulty, 3 is made up of Conditions, and 2 is Cover.
    - If the PC rolls a 8 with a Fail Effect of 5, we reduce the Fail effect first with Conditions of 3. The remaining -2 Fail Effect, would have resulted on 2 damage rolls from the Adversary or Hazard. The Cover takes these two damage rolls.
- ***The Third Priority DM is Skill.*** Skills are invoked to reduce the degree of Fail Effect only after the First and Second Priority DMs have been applied.
  - Example: The PC rolls to attack with 2d6+4 against a 13 Difficulty.
    - The PC has the following DMs:
      - Other Conditions: 3
      - Cover: 2
      - Skills: 2
    - If the PC rolls a 6 that would give them a Fail Effect with a degree of 7.
    - The First Priority DMs, Conditions, are applied to reduce the degree of the Fail Effect to 4.
    - This would mean that the PC would incur 4 damage rolls but fortunately they still have the Second Priority DM, Cover, to fall back upon which reduces the degree of the Fail Effect to 2.
    - This however would mean that the PC would still be subject to 2 damage rolls but they have the Third Priority DMs, Skills, to save them.
    - The degree of the Fail Effect is reduced to 0 which makes it an Ordinary Failure.

Fail Effect and Damage to Protection

- In Combat, reduce the Fail Effect by the PC's Skill when determining Damage rolls made against the PC.
- Take note of any Cover or Protection that modifies the Difficulty - that reduces the Fail Effect, that Cover or Protects takes the Damage.
  - Example: Medium Shields, Large Shield, Hollow or Cinder Blocks that provide Cover from an

Attack take the Damage of that Attack, up to the Difficulty of the cover or protection.
  - Assume that a PC who got a Fail Effect with a degree of 3 is taking cover behind a Shield or a stack of Cinder Blocks. This means that the 3 damage rolls that should have been applied to the PC are applied instead to the Shield or Cinder Blocks. So if the attack does 3d6 damage per attack, all of these would be applied to the Shield or the Cinder Blocks.
  - Now if the Shield that the PC had been using had a Difficulty of only 2, that means that the Shield would take two attacks of 3d6 damage while the PC incurs one attack of 3d6 damage.
  - This rule also applies to Adversaries.

Costly Failure

Costly Failure costs the PC more Time, Materials, Resources, and Opportunities. The Referee determines how high the cost is and what it entails based on the degree of the Fail Effect.

- The PC can try to determine if certain Actions lead to Costly Failures.

Success Effect

When the Dice Roll is equal to or greater than the Difficulty, the Check results in a Success Effect. The degree of the Success Effect is equal to the Dice Roll minus the Difficulty.

- See also Dice Mechanic Diagram.
- In Combat the degree of the Success Effect determines the Number of Damage Rolls.
- See Fail Effect and Damage to Protection..

Costly Success

Ask the Referee if it's Possible for this Action and Option.

A Costly Success Happens when there is a Failure, but the Character Choose to pay any price for success.

- Double the Effect of Failure and add 3, and gain a Success Effect of 0.
- This is common in Melee Combat, where a Character may fail by a margin of -2, missing, but decides to Hit the other character regardless of the cost. Suffering a -2 x 2 + 3 or -7 Fail Effect.
  - This Option makes sense if there is not much left to lose.
  - The Referee may disallow this if they judge these as not applicable or irrelevant.
- See Also Reckless Action

Advantage and Disadvantage

The Referee may rule that a PC has an Advantage or Disadvantage to their roll. Anydice link to the odds of the mechanic.

Advantage and Disadvantage rolls cancel each other out on a one to one basis..

Advantage Roll

Roll 3d6 and keep the two highest results, then add modifiers.

Disadvantage Roll

Roll 3d6 and keep the two lowest results, then add modifiers.

Time and Checks

Assume the Median Action Duration roll, and use the Success or Fail effect to multiply its smallest Unit, rolling the lowest duration unit, the reduction of time in a Success or the added time on a Failure.

Example a 1d6 second action:

- On a Success Effect of 0 is 3 seconds.
- On a Success Effect of 1 is 2 seconds.
- On a Fail effect of -1 is 4 seconds,
- on a Fail effect of three 6 seconds.

Skill Checks

Skills add to Characteristic Modifiers to Roll.

- In combat Skills are the last Defense Modifier if they are Relevant.
- Adversary Skills are part of Conditions that Modify Difficulty. A more skilled Adversary.

Stealth Rolls

Stealths rolls can be made with Recon or Streetwise Skill.

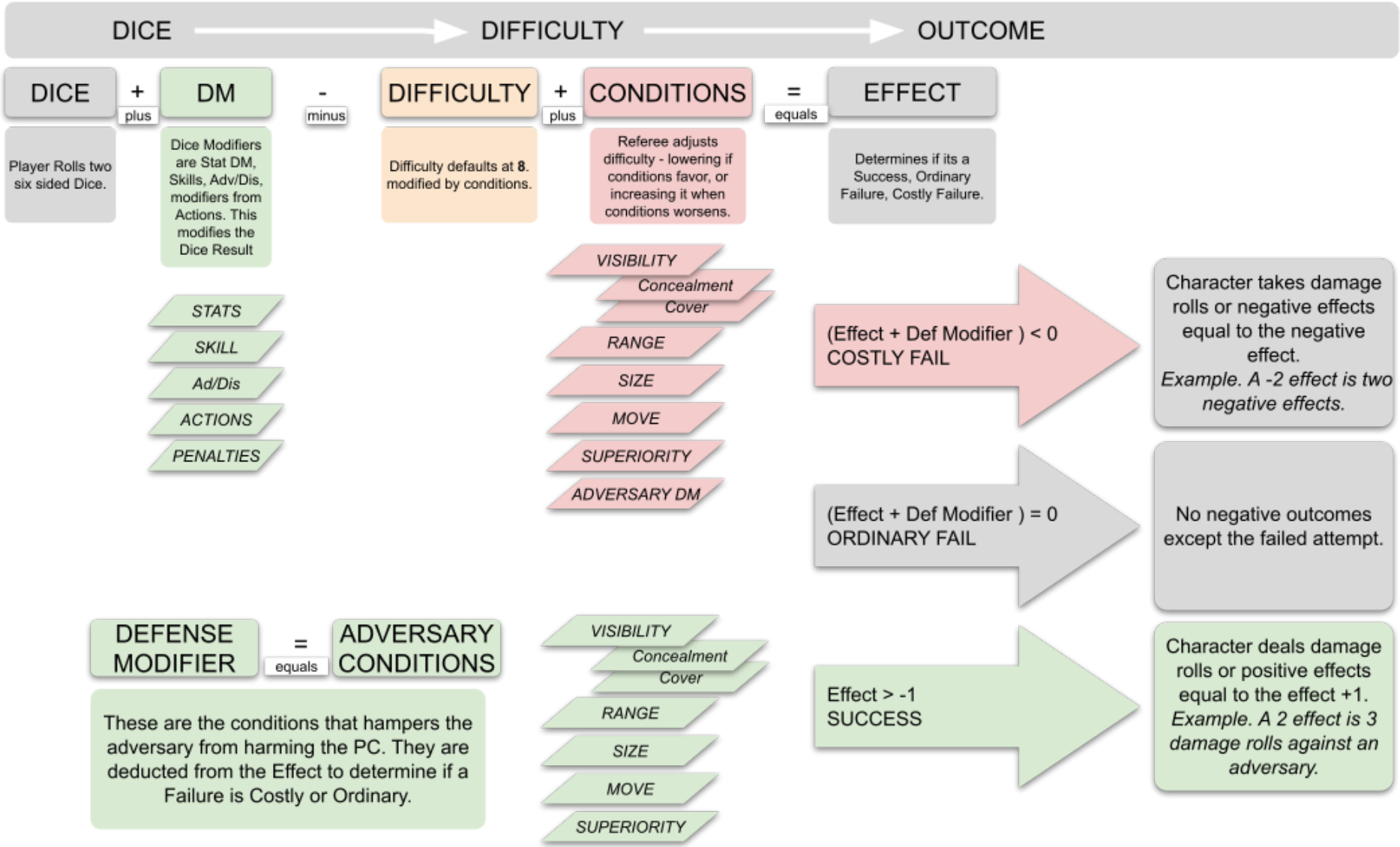
- The Referee can allow dedicated Stealth skill by allowing swapping out Recon or Streetwise with Stealth.
- Stealth also covers concealing and evading detection.

Combined Difficulties

Combined Difficulties are conditions where two or more skills are factored to success or failure. When this is the case we increase the difficulty by the additional number of skills the character doesnt have. The Referee works with the Player how the skills will be used to determine which is the most relevant.

- Example.
  - Athletic Attack - either Jumping attack, charging attack, and retreating attack.
    - If the Character has ChM 1, Athletics 0, and Attack Skill 0 and there are no other conditions then its Difficulty 8.
    - If the Character doesn't have athletics the Difficulty increases by 1 to 9. So the character rolls at a Competency of 1 against Difficulty 8+1.
  - The Referee relates conditions that Navigation, Stealth and Athletics are relevant - balance in broken ground.
    - If the Character has ChM 1, doesnt have Athletics and Navigation, just has stealth Skill 2, the character rolls at a Competency of 2 against Difficulty 8+2.

DICE MECHANICS DIAGRAM



RUNNING Mneme

Narrative Combat Summary

1. Describe the Clear and Present Danger. Emphasize what the PCs can immediately perceive in terms of Conditions and Actions.
2. Allow the players to ask for details and for clarifications. Provide details based on the questions of the players. You may also drop hints to encourage the players to ask for more details and clarification.
3. After the players have stopped asking for details and clarifications, ask them for the actions that their PCs will take. As each player answers clarify the stakes of the action, the difficulty, and the consequences.
4. Resolve the action and show how the outcome of the action changes the situation.
5. Move on to the next player and repeat the process.

Narrative Combat

Begin with the PCs and narrate threats and information based on what is immediate to them, and where their minds are.

Its useful to apply Occam's Razor and know that while there can be dozens or factors can influence events - events tend to boil down to only a few factors - Referees determine these factors and Players negotiate these factors with the actions they take.

Characters with low initiative may be unaware and may not have the presence of mind to see threats and dangers in their immediate surroundings - as they instinctively look in the horizon or far off.

Then proceed to narrate, in simple sketches and glimpses, expanding your descriptions outward from what they may see in their immediate surroundings. Those with tactical training and awareness will inspect quickly what is close and proceed outwards.

Mindsets in danger are all over the places, they do not process threats knowing the total of all threats and gauge it by priority - the brain really irrationally jumps at shadows. Its ok to get it Wrong and be wrong and mis-assess the entire situation - this is the fog of war.

Break down Description in Passes, you don't have to elaborate once you've gotten eye contact or body language or non-verbal cues from the player the understand. Focus on the threat - Ideate by answer "What will hurt/kill/maim me?" for the player.

As you play the motions and intent of the adversary limit only what can be seen and observed. Telegraph if the PC should know better (remember one roll per turn).

The Mneme system already is designed with the key Factors in mind: CONDITIONS then ACTIONS. Think in the conditions as you narrate to the players or referee. Use the list of Conditions Awareness, what all parties know, Terrain/Surroundings, Range, Concealment/Visibility, Cover, and when things can be seen Sizes and Movement. If you can mentally run this list of things you can easily narrate the threat to both players or as a player narrate your actions with impact.

With the dressing of the Conditions established, focus on a scene-by-scene narrative of what is happening to each player by the initiative.

Take note the Superiority Mechanic simplifies many Adversaries and Actions into a Modifier to the Difficulty.

Players will interrupt and will clarify and it can all be too overwhelming, but if ever you are overwhelmed just take notes and refer back to the notes. A big help is from the players sorting each other out and simply listening to their assessment. Players reiterating the situation for their understanding is one of the most efficient and proactive ways they help everyone get their bearings.

When you start getting everyone's Action you will be interrupted and people will clarify, and it's all ok. Bring focus back to the Scene with this PC. Begin again with Threats and Conditions. Elaborate on the threats with the Action or the implication of the Adversary movement.

To clearly communicate to the PC what their action is ask them for their action. Uncertainty and confusion comes from understanding how the threat relates to them - you can buy time and patience by having another player reiterate while you review your own notes.

Clarify Actions so that you can adjust the Key Factors: PC Conditions and Adversary Conditions. Reiterate the PC action for clarification - as well as possible outcome of damage and harm. Then you can ask them to roll against the difficulty you assign.

Review the outcomes, and play them out. Outcomes also change Adversary behavior and options. Adjust these as you proceed to the next PC.

Adversary.
Throughout this book Adversary is used to describe a target or any opposing force, group, or factor.

Obstacles
Obstacles are barriers between the PC's and their objectives.

Adversary Conditions
Adversary conditions affect Fail Effect of the PC and the Challenges, Obstacles, and Difficulty Adversaries have to overcome. Strategically actions taken by the PCs try to make the Adversary's Conditions worse - by taking cover, reducing their profile, moving decisively, and taking more advantageous positions.

PC Conditions
PC conditions are the conditions that typically affect the Difficulty. Adversaries will seek to reduce their adverse conditions and create more adverse conditions for the PCs.

Resources
Resources refer to manpower, soldiers, support personnel, materials, equipment, and supplies. As well as the support and logistics to deploy, maintain, and replenish this manpower.

- While Resources are not exclusively military, many of the principles discussed apply to projects and operations used by organizations, entities, corporations, businesses, and even collectives and communities.

Forces
Forces are Resources Deployed or in use. While Resources are in reserve or in a less active state.

- Forces refer to combat and military applications of these Resources.

3 Key Factors
3 Key Factors are used to predict Outcomes. The 2 key factors are used in Narrating Combat.

- **PC Conditions** - Conditions making things difficult to PCs.
- **Adversary Conditions** - Conditions making things difficult for Adversaries.
- **PC Dice Modifier**.
- Basic Key Performance Indicators. These are the values to track.
  - **Adversary Conditions**. The Degree of Allowable Fail Effect, before the PC takes damage.
  - **Dice Face to Succeed**. PC Conditions minus PC Dice Modifier plus 8. Anything greater than 12 is an automatic fail.
  - **Dice Face resulting in Injury**. Dice Face to Succeed minus **Adversary Conditions** minu 1.
- https://tinyurl.com/3yn3pbna access to a spreadsheet that allows the Referee and Player see the outcomes.

	Dice	DM	Die Result	Adversary Condtions	ΣPC Conditions	ΣDiff	Effect	Outcome	Defense Modifier	Outcome	Dmg Rolls
Player	12	3	15	2	10	18	-3	Fail	-1	Takes Dmg	1
100%	2	4	6	4	4	**12**	-6	Fail	-2	Takes Dmg	2
97%	3	4	7	4	4	**12**	-5	Fail	-1	Takes Dmg	1
92%	4	4	8	4	4	**12**	-4	Fail	0	Ordinary Fail	None
83%	5	4	9	4	4	**12**	-3	Fail	0	Ordinary Fail	None
72%	6	4	10	4	4	**12**	-2	Fail	0	Ordinary Fail	None
58%	7	4	11	4	4	**12**	-1	Fail	0	Ordinary Fail	None
42%	8	4	12	4	4	**12**	0	Success	0	Deals Dmg	1
28%	9	4	13	4	4	**12**	1	Success	0	Deals Dmg	2
17%	10	4	14	4	4	**12**	2	Success	0	Deals Dmg	3
8%	11	4	15	4	4	**12**	3	Success	0	Deals Dmg	4
3%	12	4	16	4	4	**12**	4	Success	0	Deals Dmg	5

PC Conditions: Obstacles to Hit the Adversary (Bad for PC)

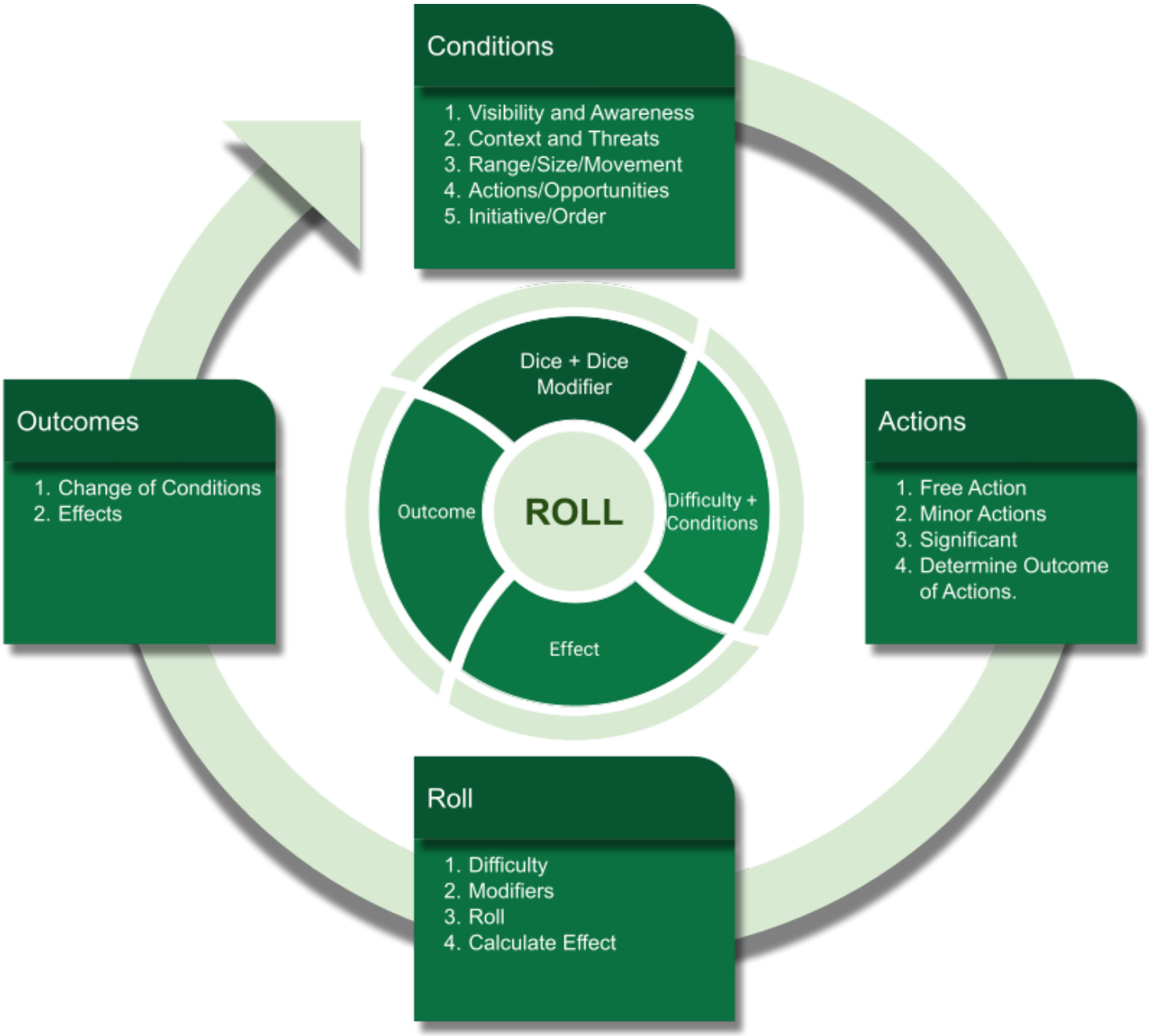
Adverasry Condtions: Obstacles to Hit the PC (Good for PC)

What Can you Know from the **PC Dice Modifier**, **PC Conditions**, **Adversary Condtions**?

8	**Dice Face Needed:** PC Conditions minus PC DM +8: how much they have to roll to Hit. Example Rolling 8 + 3 = 11 to Hit. (See Green bands). **Lower the better**.
4	Adversary Conditions: Fail Effect that Results in Ordinary Fail. **The larger the better.**
3	**Dice Face taking Damage** (Dice Face Needed minus Adversary Conditions **minus 1**). **The Lower the better.**
0	PC Conditions minus Adversary Condtions. **The lower the better**.

COMBAT CYCLE Diagram

This cycle draws from the Deming Cycle of Plan-Do-Check-Act, we can apply this cycle to each player's turn in the combat encounter. As certain Conditions become known to all the players and everyone listens to the other players act, then fewer and fewer Outcomes and Changes need to be communicated every turn.



CONDITIONS

Conditions are Difficulty Modifiers.

At any given time during the game session, the Referee and the players need to be mindful of the Conditions that frame the situation that they are currently playing out, be it from being aware that they have their backs to the door, or if it looks like it is threatening, to Rain, to how soon it will be dark, to knowing where the Exits are, or being able to find Hard Cover.

This is especially important if that situation is a Combat Encounter.

Included under Conditions are Awareness, Information, Terrain, Visibility, Range, Sizes, Movement, and Cover factors that affect the Rolls and the Difficulty.

To help break down the complexity of running the combat Mneme has four phases in its round.

Phase One: Lay out the Conditions.

This is a Narrative or Storytelling Technique where the Referee makes it a habit of describing conditions and updating them, and the Players make it a habit to ask for the current status of the Conditions that the Referee has laid out..

Phase Two: Determine Actions

Now that the Players know the Conditions, let them declare their actions.

Phase Three: Roll

Roll means resolve actions. With Conditions determined and Actions declared, there is enough for the Referee to determine the Difficulty of the Roll and the potential outcomes.

Phase Four: Outcomes

Actions have outcomes, remember that this changes conditions. Referee doesn't have to reiterate the conditions, let the Players recount them to know if there is a difference in understanding.

CONDITIONS TABLE:

CONDITIONS Type	Description	Effect on Resolution*	Notes
Awareness	PC is unaware	Disadvantage to the roll	When the PCs are being ambushed, unprepared, and unable to properly defend themselves.
Awareness	Adversary is unaware	Advantage to the roll	When the Adversaries are being ambushed, unprepared, and unable to properly defend themselves.
Visibility	Invisible Threat	Disadvantage to the roll	When the source of the threat, attacks, and danger cannot be identified and thus adequately defended against. d. The PC cannot guard from, anticipate, know, or defend from the attack because they do not know where it is coming from.
Visibility	Attacking while Invisible	Advantage to the roll	The Adversary cannot guard from, anticipate, know, or defend from the attack because they do not know where it is coming from .
Visibility	Directly Seen	Modifiers to Difficulty from Range, Size, and Move Conditions	Can attack or target the PC or Adversary.
Visibility	Indirect Knowledge	Concealment, Cover, Range Influences Difficulty	the character or adversary has to use Recon (to observe) or Tactics to infer to verify if the character is still there. Can attack the last known area or approximation of the PC or

			Adversary. Adversary may not be there.
Visibility	Partially Concealed	+1 Difficulty	When a third or more of the Targets profile is Concealed Basic Camouflage Mist or High Humidity.
Visibility	Mostly Concealed	+2 Difficulty	When half or more of the Targets profile is Concealed Examples: Mostly Dark, Dense Mist or Light Fog, Excellent Camouflage.
Visibility	Fully Concealed	+4 Difficulty	When a Targets profile is fully concealed. Example. Complete Darkness, Behind Fog
Visibility	Partially Covered	+1 Difficulty	When a third or more of the Targets profile has cover A medium shield in Guard position would count as providing the Partially Covered Condition. Example: Foliage, Urban ruins, Sparse structures
Visibility	Mostly Covered	+2 Difficulty	When half or more of the Targets profile has cover. A large shield in Guard position would count as providing the Mostly Covered Condition. The PC crouching behind a medium shield would provide them the Mostly Covered Condition. Examples: Jungle, Thick Foliage, Trenches, Reinforced (Sandbags) Positions.
Visibility	Fully Covered	+4 Difficulty	When a Target's profile is fully behind cover. Examples: Crouching behind a large shield would provide this condition but limits the kind of attacks a Target can make, PIllbox, Deep Trenches, Behind walls and structures.
Terrain	Gravity	Disadvantage without proper skills and tools	Characters can use Zero-G or Athletics skill to prevent them being out of control. Disadvantage when they don't have Sufficient Zero-G or Athletics Skill.
Combat Conditions	Ranged in Close Quarters	+2 Difficulty	Firing range weapons within reach of a character parrying or striking the attack to defend themselves - a Defense modifier to the defender and a Task Difficulty modifier to the Attack (increasing the Difficulty).
Superiority		See Table	Modifies the condition when one party has more Time, Manpower, and Resources.

Awareness of Threats

PC is Unaware of Threats

Characters who attacked while unprepared, unaware, ambushed have a Disadvantage.

Adversary is Unaware

Characters who are attacking an unprepared, unaware, and successfully ambushing their Adversary have Advantage.

Time, Terrain and Weather

Determines Visibility, Range, and Movement. Time, Terrain, and Weather.

- Use Concealment to determine the Visibility in Fog, Mist, Rain, Storms.

- Extreme heat and cold affect the exhaustion of the Characters, dealing 1 exhaustion damage per hour or per day.
- Check the table to movement penalties.
- The Terrain Depends on the Scale of Combat: from man to man, to squads, armies, vehicles, or Systems.

Terrain Table

Terrain is about the Physical Characteristics of the Surroundings. Any running or quick actions that doesn't check for hazards trigger rolls.

Terrain	Conceal-ment	Cover	Move	Notes
Open Ground	None	None	None	Physical Terrain that has no cover, concealment, and can be moved through with some ease.
Broken Ground	for crouching or prone	for crouching or prone	1/2	Roll Athletics Dex when moving more than 12m per round. ● Ordinary Failure is falling or stopping (with no cover) ● Uneven Terrain may provide cover or concealment to a prone or lying
Light Foliage	Partially	Partially	1/2	Managed Wilderness where the trees are spaced apart or that is cleared by wildfires over time.
Heavy Foliage	Mostly	Mostly	1/2	Unmanaged and choked wilderness, sometimes impassable as the floor is crowded with competing plants.
Ankle deep Water			1/2	Its possible to run but this is Hazardous because dirty water hides hazards.
Knee deep water	for prone	for prone	1/4	same as ankle deep but murky water can hide much more.
Chest deep water	for crouching	for crouching	swim	as above but wading is difficult at one space per round.
45 degree slope or greater	behind the slope	behind the slope	climb	none

Gravity

Gravity is part of the Terrain or the Physical Conditions of the surroundings. Micro-Gravity or Zero-G causes disadvantage in actions.

- Characters can use Zero-G or Athletics skills to prevent them being out of control.
  - But they have to limit their Combat Skill by their Zero-G or Athletics skill.
  - But they still need something to anchor against. Thruster packs, Magnetic Grapples, Lanyards, or other characters physically aiding them help in this if available.
  - If they don't reduce their equal to Zero-G or Athletics skill, they will suffer Disadvantage to the roll.
- Firearms without the Recoilless property have disadvantage to attack rolls and free fall without any anchoring or leverage.
  - Firearms with Recoilless do not suffer Disadvantage in Zero or Low Gravity.
- Combat Thruster Packs have software to offset recoil at the expense of Delta-V.

Time Scale

When assessing the Conditions the Referee should consider the Ranges and the number of Characters to determine the Time Scale.

Combat Rounds
A combat round is about 6 seconds. Combat rounds are appropriate for: 300 meter or less engagements.

- 300 meter or less engagements of people fighting on foot.
  - 300 meters assume two parties 300m appart moving to engage and closing a distance with weapons that engage 20-100m.
  - up to 500 meters on mounted combat. Mounts that move +30m per round (20kph) and typically engage with weapons effective up to 20-100m.
- Smaller than 3 Parties Engaged or 50 or less people.

When a lot of the action depend on individual actions over coordination, Combat Round scale is the best.

Minutes
Minutes scale happens when there are

- Ranges are up to 500m on foot or 1km mounted on beasts capable of sustained +20kph speeds, moving 30+ meters per round or 5m per second.
- Groups of over 50, but less than 300 trying to coordinate, engage, ambush, or head off each other.
- Social Conflicts and Engagements.

dMinutes
This is the time scale for bands or battalions (groups of 300-500 combatants) fighting and maneuvering over a 5-10 KM, vehicular pursuits of up to a hundred KM, or space vessels engaging in thousands of kilometers.

Hours
This is the time scale for intense military engagement Brigades and Divisions of men, of fleets in persistent hunts or harrying attacks, or space vessels with their dance of Electronic Warfare and positioning.

Half-Shifts
This is the scale of Task Forces, and Field Armies. This is the scale where vessels maneuver for strategic advantage before they erupt in sudden violence.

Days
This is the scale of Logistics, Strategy, Diplomatic Maneuvering, and Preparation.

Visibility
The Referee and Players have to keep track of and assess the Visibility Conditions. Visibility also refers to the level of Concealment.

Spotters
Spotters are those who are viewing the Target. Spotter and Target.

Target
Target refers to what is either Visible or Concealed. this can be a PC, Adversary, Object, Vehicle, etc…

- Size Modifies Visibility, the Larger the target it, the easier to see (lowers the difficulty) and the Smaller the Target the harder to see over a distance (increases the difficulty)
- Target's stance, position , and use of surroundings affects Visibility..
- Range Modifies Visibility: the distance of the Target to the spotter affects Visibility as well as the spotter's access to tools such as optics, and sensors..
- Blindness or Poor Vision also affect Visibility and Concealment. A nearsighted person without glasses can may be considered blind or their targets are having the benefit of concealment.
- Target will be used to describe the object of a Character's actions.

Directly See
o Directly See' means that a Spotter is able to independently see, verify, and know information, position of an adversary, goal, target etc.

- Cover and Concealment that is less than Full or whose Conditionss do not add up to 4 are conditions that still allow a Target to be Directly Seen by a Spotter..
- Once a Target is Fully behind Cover or under Concealment, then the Spotter has to resort to using Indirect Knowledge to determine if the Target is still there.
- Synonymous to "***Within Line-of-Sight***"
- See Also: Blind, Invisible Threat, Attacking while Invisible,

Indirect Knowledge
When a Target is fully Concealed or Covered, the Spotter must resort to Indirect Knowledge to determine if the Target is still at the location being scanned.

- A Spotter attacking the location where a Target is known to be still incurs the Concealment Penalty.

When a Character benefits from Concealment, they may use this is as a Defense modifier.
- Other Skills and Actions are used to Indirectly know if searching nearby clues, Example:
  - Recon is used to look for clues and traces of where the Target can be, or have gone through.
  - Tactics by process of elimination deducing where the Target would be based on previous conditions and actions taken.
- Spotters who attack Targets that they cannot Directly See may still do so if they gain Indirect Knowledge of the presence of the Target.
  - For example, a Target who has gone behind a tree to prevent a Spotter from Directly Seeing them can still be attacked by the Spotter if the Spotter is able to get Indirect Knowledge of the presence of the Target's such as seeing the Target's shadow, seeing cigarette smoke from the Target, and so on..
- See Also: Blind, Invisible Threat, Attacking while Invisible,

Invisible Threat
- An Attacker is considered an Invisible Threat when the Target of the Attacker cannot identify the source of the threat, attack, or danger or the direction of its origin and thus is unable to properly defend against the attack..
- The Adversary cannot guard, anticipate, know, or defend from the direction of attack.
- See Also: Blind, Indirectly Know, Directly See

Attacking while Invisible
- The Adversary cannot guard, anticipate, know, or defend from the direction of attack.
- See Also: Blind, Indirectly Know, Directly See

Specific Cases Regarding Visibility

Optics
Are any type of augmentation the visual abilities of characters.

Depth Perception.
Using Monitors or Cameras removes Depth Perception and adds 2 to the TRN of any action where knowing range and depth information is critical. Typically this affects Melee and Ranged attacks.
- Being forced to use only one eye removes Depth Perception
- Without depth perception, combat conditions or a dangerous environment is more uncertain and frustrating. The inability to determine the distances of threats plays to character fears. There is also the fear that arises knowing only one eye remains from being blind.

Night Vision
humans actually have pretty good night vision but it takes them about 3 minutes to adjust. A character who has not adapted to the dark suffers a -2 to their Conditions
- It takes 1d3 rounds for a human to adjust to a Sudden Bright Light. Before the adjustment, the human is temporarily blind..

One-Eyed Night vision.
You can keep one eye covered so that it automatically adjusts to darkness which allows Spot and Navigate skill tests in the darkness. However, since only one eye can see in the dark, you lose Depth Perception.

Red Light
Red light disrupts night vision minimally and is not visible from a distance.

Directional lights
Directional reduce darkness penalties of 1-4 for certain ranges in ranges of up to 10m, 20m or in extremes of 50m.
- Directional lights will have a Range and a Level rating of 1-4. the level rating reduces Darkness modifiers by such.
  - Every Double the Flashlights Range, reduces the Level effectivity by half. Less than 1 is zero.
- Example.
  - Hooded Lamps 2m L4. 4m L2, 8m L1
  - Bullseye Lanterns 5m L4, 10m L2, 20m L1.
  - Flashlight 5m L4. 10m L2, 20m L1.
  - Flashlight 7m L4. 14m L2, 28m L1.

Spotlights and SearchLights.
These are Flashlights scaled up and used by Vehicles. 50m and up.
- Spot lights 50m L4, 100m L2, 200m L1,
- Search Lights 100m L4. 200m L2. 400m L1.
- Search Lights 200m L4. 400m L2. 800m L1.

Scopes
Telescopes, Binoculars, Cameras that have magnification abilities. This negates or reduces some Range Condition modifiers to vision rolls.

Infrared/Heat Vision.
Be able to see Heat or Infrared. Windows may allow visible light but block infrared light.

One-Eyed Infrared/Heat Vision.
Sacrifices depth perception to be able to see heat profiles and signatures.

Light Intensifier
Light Intensifiers have levels, and it reduces Darkness Penalties equal to its level. Darkness penalties range from 1-4.
- Bright Light temporarily blinds 1d3 rounds to adjust.

One-Eyed Light Intensifier
Negates Darkness Penalties, but suffer Difficulty 2 when needing Depth Perception.

Heads up Display
At TL9 some visual feeds can be overlaid on the character's view. Information fed to the HUD is a free action to take in, instead of a Minor Action.

Sensors
Sensors do not enhance the Visual abilities of the characters but grant additional information that aid in giving context.
Example: the character may not see a threat, but they can hear it, smell it, feel its heavy wheeled rumbling,

Protected and Augmented Listening
Protects against sound attacks, and grants a +1 DM to Recon Listening checks.

Vibration
Vibration Sensors attempt to triangulate a source of vibration. It can only give Directions of sources and it needs to be able to .

Environment Visibility
This affects the ability to see directly or indirectly the adversary party's actions. A threat, very far off beyond the range anyone can do anything about it.
- One can see 2-5km on open terrain, until a terrain feature limits that visibility.
  - Distance to horizon, without atmospheric limitations, is d (distance in kilometers) equals the square-root of height in meters times 3.57
    - $d \approx 3.57\sqrt{h}\,,$

Example	Height	Distance
Person	1.75 m	4.7 km
Low Wall	2.00 m	5.0 km
House Roof	3.00 m	6.2 km
A tree	4.00 m	7.1 km
Tower or Castle Wall	10.00 m	11.3 km
Tower on a Hill	20.00 m	16.0 km
Mountain Watch Tower	500.00 m	79.8 km

Concealment
Concealment is anything that reduces the visibility of targety. It can be the weather, shadows, smoke, darkness, sensor jamming, sensor or visual noise that overwhelm the ability to discern and identify. See Visibility.
- Concealed and Cover stack but cannot exceed value of 4 (which is completely obscured). A Target can benefit from Partially Covered and Partially Concealed for a Conditions of 2, but cannot benefit from completely obscured and covered.
- Basic Camouflage provides Partially Concealed - like Olive Drab greens, White for winter, or khaki for desert terrains.
- Advanced Camouflage provides Mostly Concealment on their specific terrain - Flecktarn, Cadpat, Marpat, Multicam, Digital Camo, Woodland, etc…
- Sensor Invisibility. Cloaking technology, Electro-magnetic absorption and other such technologies can be used to become invisible to sensors.
  - Heat or Infrared Invisibility. Target blending into the background heat or infrared.

Cover
Cover is a Physical barrier to the Target. Often what is a physical barrier is also a visual barrier, but characters can take cover from something like bullet proof glass, a material that gives away their position or presence (like their shadows being visible on its surface).

Cover takes Damage

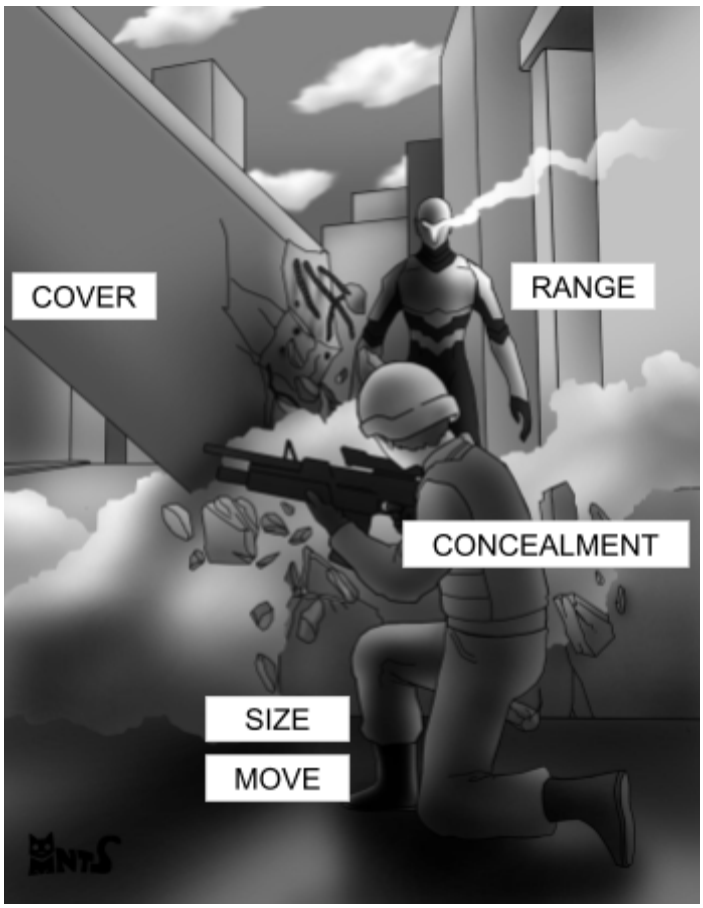
On a margin of Failure - both parties who benefit from cover or protection have that protection whittled down.
*Indirectly knowing, penetrating cover.

- See Damage to Protection

Range, Size, Move

The referee can apply relevant Range, Size, and Move Difficulty or Dice Modifiers.

- Range Modifier affects Vision and Ranged Attacks. the farther something is the more difficult it is to distinguish, target, or assess it.
  - This measures ranges in Meters, but Scales to KM and can scale up to Light Seconds, Minutes, and Astronomical Units. In these scales equipment, vehicles, or ships offset these modifiers. Example
    - Binoculars and Scopes grant a Vision or Aim modifier of +4 to +8. this offsets penalties from Range Size Move. Example: searching for something 100m away has a +4 difficulty, adding up to 12 difficulty. Binoculars rated for 100m would grant equipment modifier of +4 to the roll.
    - Aerial Recon have Vision Conditions in kilometers range (Conditions 8 and up)
    - Spaceships have sensor systems at fractions of 1 light second increments may have +26 modifier to compensate for Ranges of a light Second far.
- Size Modifier affected by
  - Character Position: Prone, Lying, Crouching, or standing to full height unaware.
  - the Characters Size or the Object's Size.
- Referee Can Apply Range Size Move Modifier to Attacks and Visions Rolls when applicable.
  - On Vision and Ranged Checks it depends on the Terrain.
  - Range and Move don't usually add up.



Range Calculations in Spaces.

From the center of Character's Space to the Next Space.

Adjacent Space

Adjacent Space is the spaces right beside and in-contact with the Character's space.

Reach 1 Space

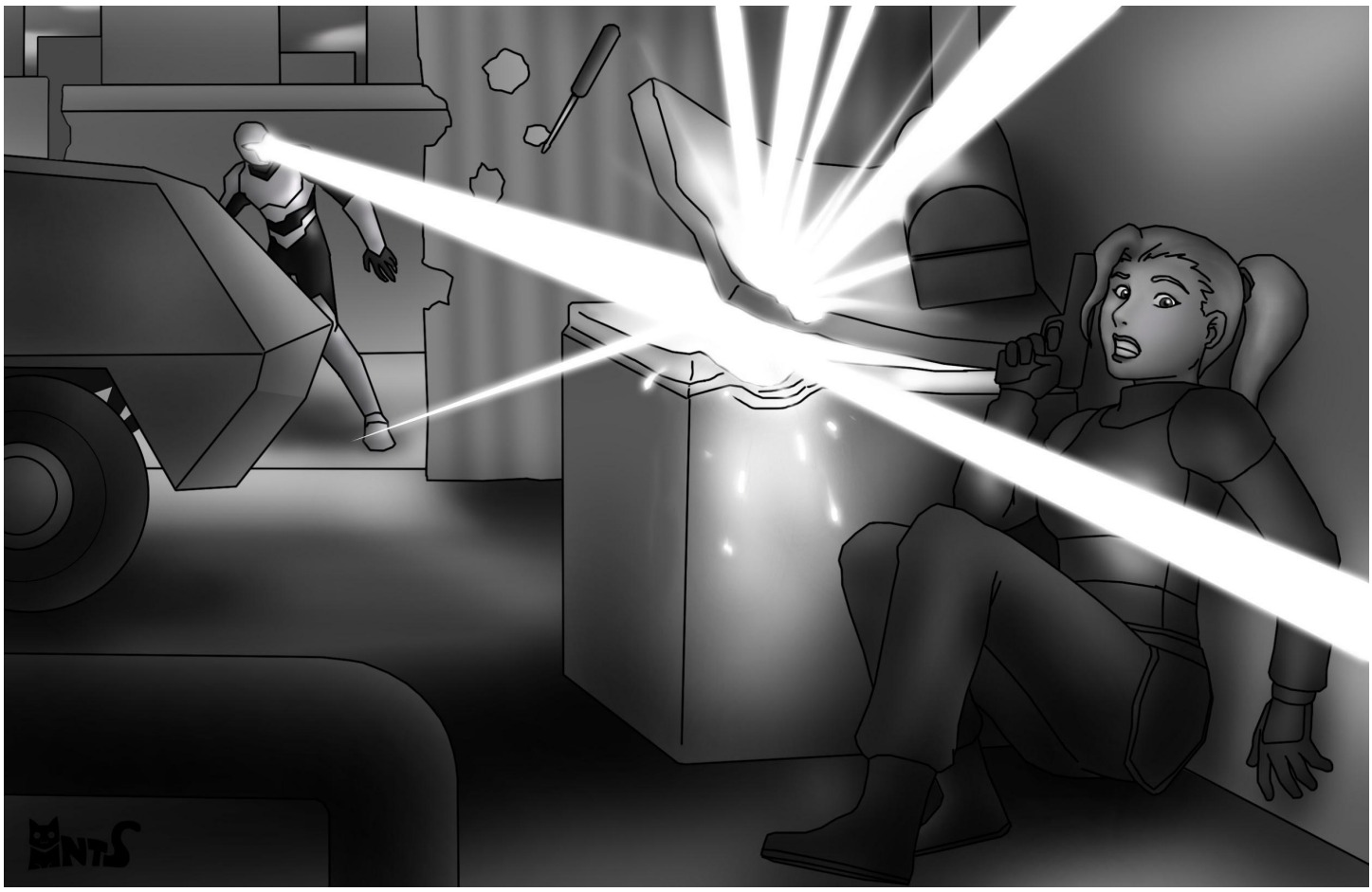
1 Spaces Away from adjacent. 2 spaces from the character's space.

Reach 2 Space

2 Spaces Away from adjacent. 2 spaces from the character's space.

Personal Space

In the character's space, daggers, grappling and unarmed attacks can be used in personal space. Larger weapons are unusable, and in some cases at a disadvantage.



Range Size Move Table:

Range/ Size	Range Mod	Size Mod	Move Mod	Kph
0.5m	Melee/-3	+2	0	1.8kph
1m	Melee/-3	+1	0	3.6 kph
1.5*	Melee/-3	0	0	5.4 kph
3m	-2	-1	+1	10.8 kph
5m	-1	-2	+2	18.0 kph
10m	0	-3	+3	36.0 kph
15m	+1	-4	+4	54.0 kph
30m	+2	-5	+5	108.0 kph
50m	+3	-6	+6	180.0 kph
100m	+4	-7	+7	360.0 kph
150m	+5	-8	+8	540.0 kph
300m	+6	-9	+9	1080.0 kph
500m	+7	-10	+10	1800.0 kph
1km	+8	-11	+11	3,600.0 kph

*Adjacent space (square or hexes).
see Move Table

How to use the Range Size and move Table.

*This is also Meters per Second.The KPH, or kilometers per second, is the conversion of the meters per second.

This table is used to check the Conditions affecting the Target or Adversary. Remember that Conditions are the domain of the Referee and as such modify the Difficulty. The plus or minus of these factors considers how they affect the difficulty.

Lets go by the order of Range, Size, and Move.

First, check the Range and the Range Modifier. Adversaries farther away are more Difficult to affect. If the Value of the Range/Size/ mps is equal or greater than Target/Adversaries conditions.
If the Adversary is 2 meters away, the Range modifier is -3 difficulty modifier. Take note that the default Difficulty is 8 (which makes it a 5).

Next is checking its Size, If the subject is Man sized, less than 3m in size like 1.75m, then there are no modifiers. If its Larger, like a car, the difficulty is -1.

Next is checking its movement. The range and size column is also the meters if its movement is pretty slow, less than 3m/s or 10kph then there are no difficulty modifiers.
Example
A target or adversary that is:

- 15m away (+1), Less than 1.5m example crouching, (+1), moving at 20kph (+2) would have difficulty of 8 (default) plus 1 plus 1 plus 2 for a total of 12 Difficulty.
- 100m away (+4), Less than 0.5m prone (+2), lying still (+0) would have difficulty of 8 (default) plus 1 plus 1 plus 2 for a total of 14 Difficulty.

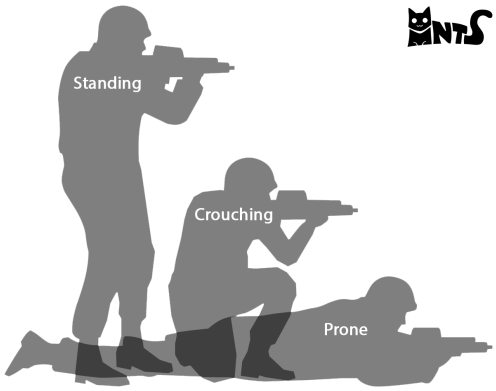
This table also is used in the Inverse, what conditions are affecting the Adversary from affecting the PC.

Sizes, Positions, and Stances

The Position of the Target determines how much of a Profile or Overall Target the character is for an attack. The larger the Profile, the easier is to be attacked. The smaller the profile, the more immobile and uncomfortable the stance.

Stance	Profile	Size Mod	Move	Defense
Standing	1.5m	0	Full	Full
Crouched	1m	1	1/2	Ad/Dis*
Prone Crawling	0.5m	2	1/4	helpless
Prone Flat	0.3m	3	None	helpless

*if Attacking an crouching adversary gain Advantage.
*if crouching against a melee adversary roll at Disadvantage.



Superiority Modifier

Instead of having to track a myriad of modifiers from components such as conditions, actions, equipment, resources, manpower, materials, and so on, the Referee simply takes all of these into account and determines which side has Superiority in the encounter.
When the PC has superiority reduce the difficulty (a negative condition modifier), when the Adversary has superiority increase the difficulty..

Superiority Table

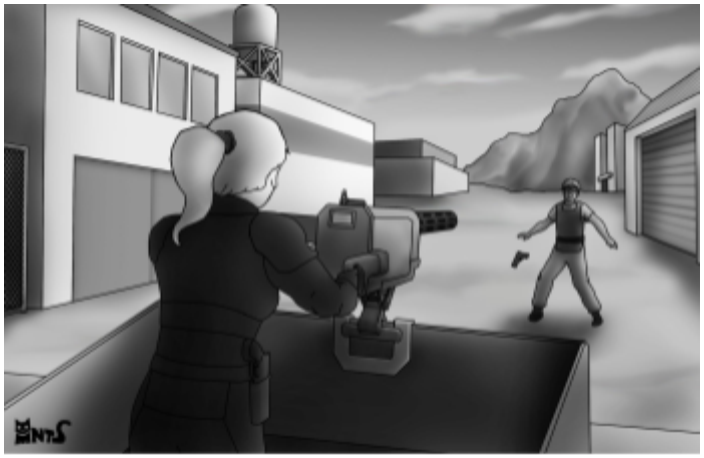
Superiority	Difficulty Modifier	Examples
1:1	0	Equal Numbers.
3:2 1:1.5	1	3 Automatic Rifles vs an Assault Rifle.
1:2	2	Machine Gun against Assault Rifle.
1:3	3	A soldier with a suite of drones, or a warrior with a trained pack of wolves.
1:5	5	A fire team against one ranger.
1:7	7	A spear saint against the picked men of a clan.
1:10	10	A cohort of soldiers against a Legion

- This is not Limited to just Combat. Superiority applies to PC or Adversaries with more resources, time, manpower, than the PCs in social, professional, or technical Tasks. Up to what is relevant.
  - Racing to a Destination with Superior resources grants a bonus.
  - when the adversary has a Team of 10 in a project and the PCs are only 3, that's a 1:3 or +3 Difficulty.
- This applies to Action Economy or Time, PCs or Adversaries that have more Time and thus more actions may apply Superiority Modifier to the Difficulty
- When PC's or Adversaries Aid One another the Referee can apply Superiority as well to simplify.

Superiority in Numbers

Superior Firepower

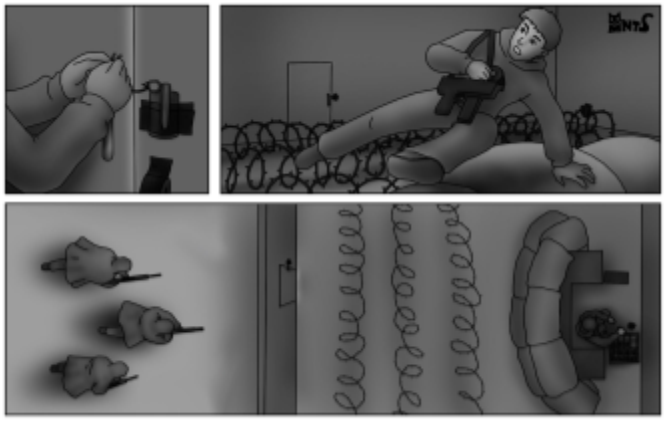
- When one party has more Total Volume of Fire than Another Party.
- Example.
  - Automatic weapon vs Auto 2 weapon have a 1 vs 2 Superiority Modifier.
    - A Partisan with a Automatic rifle against an Infantryman with a submachine gun.
    - A Policeman with a semi-automatic pistol against a bank robber with a Submachine gun.
  - Auto 2 weapon vs Auto 3 Weapon have a superiority modifier of 2 vs 3.
    - A gang-banger with a Submachine gun, against soldier with an Assault Rifle.
  - Automatic weapons vs Auto 3 weapon have a superiority modifier of 1 vs 3.
    - A pistol against a machinegun.



Superior firepower

Superior Action

- Action Economy - the concept that the side with the most Actions have the advantage over another side, Superiority Modifier simplifies this as a Modifier to save time.
- Drones, Intellect Programs, Robots, Agent Programs, Bound-Spirits, and many other Intelligent or Semi-Intelligent entities able to perform actions and work for one party are simply resolved with the Superiority Modifier.
- This allows the Referee to manage…
  - player spotlight - how much air time it takes to explain a complex set of compound actions.
  - game complexity, by distilling a group by their strongest or weakest and by their resources and manpower.



setting traps, deploying resources.

The Combat Round Summary

- Each combat round lasts around six seconds of game time.
- Initiative is rolled once at the start of the Combat Encounter to determine the sequences of actions, using 2d6 plus any Characteristic Modifier and relevant skill.
  - PCs keep their Initiative Value throughout the combat.
- In a combat round each character gets a minor actios that can be used individually or as significant or full action.
- Actions are taken in descending order of Initiative.
  - If two characters have the same Initiative, the character with the highest Dexterity goes first. If they are still tied, then characters act simultaneously.
- A PC's turn is not completed until they have used all their actions..
- Once everyone has acted, the Referee declares that Combat Round over. The Referee then determines if there is a need to begin another Combat Round.

Initiative

Everything happens simultaneously in a round, but some actions, and outcomes influence other actions and outcomes. Initiative represents the order of influence.

- Characters Roll 2d6 + Relevant Skill or Modifiers depending on the skill they apply in the start of the combat. Examples are:
  - Using Recon gives the PC the conditions.
  - Tactics to know what to do or predict what the adversary will do.
  - Weapon skill to ready their weapon.
- Initiative resolved with each Player rolling 2d6 + the Relevant Characteristic Modifier or Skill they will be using for the Combat Encounter. They must perform a corresponding action on the first Turn using that skill.
  - Example: A character with Recon, Tactics, Leadership, Melee Blade, Gun Combat: slug can use these skills to determine their Initiative.
  - In Mneme Skill influences Initiative far more than Ability. With Stats being Equal, Skill counts for seizing initiative.
- If the combatants are all unprepared for combat, then each rolls 2D6 and adds their Dexterity DM to determine starting Initiative. Initiative determines the order that characters act in, but it can also be spent to react to events. If some of the combatants are ready for combat and some are not, such as in an ambush, the prepared characters are considered to get an automatic 12 on their roll, giving them an Initiative of 12 + Dexterity DM. Characters that are aware of a group, and that group is unaware of them, may elect to avoid conflict entirely.

Characters that have the Tactics skill may make a Tactics check and add the Effect of this check to the Initiative of everyone in their unit with whom they are in direct communication.

ACTIONS

Action Class. the amount of minor actions it takes in a turn.
Action Type. Categorization that group it together with other similar actions.

ACTION TABLE

ACTION Class	ACTION Type	Description	Description
Free Action		Quick Phrase	● A quick utterance, or a simple shout.
Free Action		Quick Glance	● A quick glance limited the character's forward field of vision.
Minor Action	Minor Movement	Movement/ Change stance	● Move 6m or 4 spaces under Light Load. see Move table ● Move 4.5m or 3 spaces under Medium and Heavy load. see Move table

Minor Action	Minor Readying	Grabbing/ Taking something accessible	• Drawing from Open or Quick Release pouches, pockets, holsters, bands, or other containers. • Arrows from Quivers
Minor Action	Minor Ready	Readying	• Minor Loading (mLoading)
Minor Action		Aim	• Gain a +1 DM to the roll. Maximum 6 Aim bonus.
Minor Action	Minor Action	Quick Scan	• A quick scan that can cover the 360 field of vision. • Reveals anything not using Deliberate Stealth.
Minor Action		Minor Stealth	• Adversaries not using Quick Scan are not aware of this character. • Move 3m (2 spaces).
Significant Action	Skill Action	Skill Action	• Skill Actions are uses of skills that can be performed in a significant action.
Significant Action	Skill Action	Deliberate Stealth	• Move Stealthy 6m (4 spaces), roll Against Adversary Recon • Evade detection anything less than Scan/Search (Quick Scans and Glances).
Significant Action	Skill Action	Scan/Search	• Roll Recon against the Deliberate Stealth of Adversaries • Detect anything less than Deliberate Stealth

Significant Action	Skill Action	Dash	• Move 15m on a straight line.
			  ○ This makes the PC/Adversary harder to hit.
			  ○ Roll Athletics if terrain can cause a suddens top or fall.

Significant Action	Skill Action	Use device	• Activate or Use a Device. Give commands to a Drone.
Significant Action	Skill Action	Use Tactics	• Roll tactics against chosen difficulty level, gain a number of Adv/Dis, rolled a number of times equal skill+1 per combat
Significant Action	Significant Readying	Significant Ready	• Reload, Draw ammo/mag,
Significant Action	Action OPTION	Reckless Action	• Gain a +2 DM, but even if success and after resolving success, apply a -3 to Effect. If the Effect would have been negative resolve the negative outcome.
Significant Action	Action OPTION	Supporting Action	• Roll and Success effect is a DM bonus to characters that benefit from the Support.

Significant Action	Action OPTION	Coordinating Action	• This action is considered to have resolved simultaneously with another character.
			  ○ Gain Superiority Modifier to the Roll or
			  ○ A special effect the Coordination intended

Significant Action	Action OPTION	Plan/Wait	• Predict Adversary Action, gain Advantage if plan is sound and prediction is correct or Disadvantage if prediction is wrong.
Significant Action	Action OPTION	Supporting Actions	• Aid another Character, the other character gain a DM modifier by the Success Effect.
Significant Action		Defend	• Double one Condition or Skill's value to reduce Fail effect.
Significant Action		Attack	• Attack one or more adversaries.
Significant Action	Attack OPTION	Ranged Attack	• Throwing or Shooting attack.
Significant Action	Attack OPTION	Melee Attack	• Attacking with melee weapons or unarmed strikes.
Significant Action	Attack OPTION	Setting the Rate of Attack	• Set the rates of attack, the Use of Autofire or attacking with weapons in each hand.

Significant Action	Attack OPTION	Targeted Attack	• Attack a specific Area
Significant Action	Attack OPTION	Grapple	• Gabbing and Wrestling an adversary.
Significant Action	Attack OPTION	Moving Attack	• Attacks that have an element of movement like Leaping attacks, charging attacks, tackles, bull rush or slams.
Full Action	Full Action	Skill use	• Roll Skill actions where success or fail effect determine how quickly an extended action can be accomplished.
Full Action	Full Action	Run	• Roll Athletics running through obstacles, hazards, or enemy fire.

Minute, dMin, Hour, and the expanded time scales of action will follow in future installments.

Movement Table

Each space is 1.5m. Move is one Minor Action. Dash is a Significant Action. Run is all 3 minor actions.

Scale	Action	*Light*	*Med/Hvy*	*Light Crouch*	*Med/ Hvy Crouch*	*Crawl*
1.5m	*Move*	4	3	2	1	NA
1.5m	*Dash*	10	7	5	3	2
1.5m	*Run*	15	10	7	5	3
4.5m	*Move*	1	1	NA	NA	NA
4.5m	*Dash*	3	2	2	1	NA
4.5m	*Run*	5	3	3	2	1

Actions Types

Free Actions

These are actions that do not take up any minor action, but the Referee will limit these actions when done repeatedly.

Minor Actions

These are quick tasks that take little thought or attention. These are typically so simple as almost involuntary actions or have been drilled or conditioned into the character. There are three minor actions in a round.

- 3 minor actions per round.

Significant Actions

Significant Actions takes thought and attention, which is why using skills, attacks, defense, and any deliberate action is typically a significant action.

- Significant Actions Costs 2 minor Actions.

Full Actions

Full Actions are actions that occupy the entire round that prevents the Character from doing anything else.

- Cost 3 minor Actions.
- Because of the nature of the game mechanics the Referee simply narrates what's going to happen and the Player declares if they will continue or Abort the full action.
- Characters who choose not to abort their Full Action cannot defend themselves and can be considered helpless. This grants characters attacking an advantage or automatically succeeds on attacks.
  - The Player still rolls - rolling the effectiveness of their full action or how well they can take the hurt See Helpless.
- **Let PC's who perform Full Actions Roll Every Round**.
  - They either roll for the Success Effect to speed up their progress.
  - Or they Roll against the Changing Conditions of their Task.

Action Options

- Action Options are Options that can be used to modify a significant action.
- It is possible to perform a Reckless Coordinated Supporting Action like emerging from cover under enemy fire, timed to give your allies an opportunity to maneuver. It being effective depends on the roll.

Examples of Free Actions

Some tasks take so little time to complete and require very little attention from the PC that they do not even qualify as a minor action, for example, shouting a warning, pushing a button, checking your watch, and so on.

A PC may perform as many of these free actions as they like in a turn, though if they perform so many, the Referee may require them to spend a minor or even a significant action on their various tasks.

Quick Phrase

- Any speech that can be spoken in one breath.
- Repeating phrases over and over again.
- declaring action over the radio.

Quick Glance

- A single quick glance without turning one's head

A Snap

- Pressing a detonator button already at hand, Activating a program with one press.
- Dropping an Item already Held.
  - Example dropping a charged grenade as a free Action, but what the else the character does depends.

List of Minor Actions

Minor actions are actions intended to move a distance or to manipulate or move an object. You can take up to three minor actions per round, at the loss of a significant action.

Aim

Aim allows the character to Fine Tune and adjust their ability to shoot a target.

- A character can Aim as a minor action.
  - if the weapon has an Aim Modifier, they gain the Aim modifier of the weapon on their next Attack.
  - If the weapon doesn't have any, they gain a +1 Aim Modifier to attack.
    - the PC can take up to 6 minor actions for a +6 or more (because of Weapon Aim Modifier) to their next range weapon attack.
- Aim is not Exclusively preparing to shoot something, it could be correcting shots made from seeing Laser Sights, impact, tracer rounds, etc… Ballistic Correction
  - if the Character was shooting on a previous round they can benefit with better Aim. See
- Aim can be used within the area of the line of Sight of a character. Aim can be applied to Adversaries that Appear in the area aimed.
  - Using aim in this manner is called "Covering" an area.
  - Conditions that improve Aim are:
    - Weapons Aim Modifier
    - Ballistic Correction
- **Keeping and Losing Aim Modifier**. Character's keep their Aim Modifier on the same target if the

target's conditions has changed drastically - the Referee considered the conditions more of the same.
    - If there are Multiple Targets in the same Position and Condition the Referee determines how much of the Aim Modifier is preserved.
    - Characters lose their Aim Modifier when Changing Targets in a Different Positions and Conditions.
- Example
    - A sniper takes an Aim minor action, lining up its Scope mounted rifle for a +8 modifier against a Target half a KM away (Range Modifier +7; with other Conditions of 2, a total 17 Difficulty), with Competence of 3, the total modifiers are 11.
    - PC rolls a 2d6+11 against 17 which needs a 6 or better roll to succeed. That's about 72% chance.

Minor Move

The character minor moves up to six meters or 4 spaces. When using a square grid for modelling the combat, each square should measure 1.5 meters to the side. This means that the average character moves four squares on his turn.

- 6m or Four 1.5m spaces (squares or hexes)
- Difficult terrain, such as rubble, mud or thick under-brush can Penalize a character's movement.
- Medium Load and Heavy Load - reduces move from 3 to 4 spaces.
- **Crouching** also halves movement.
- See Terrain, Dash, Running

Changing Stance

Changing stance is a minor action.
A character can change to any one of the three stances – prone, crouched or standing – as a minor action. See Size, Position, Stance

- Standing to Prone or Crouching can be done in One minor action,
- but Prone to Crouching takes 2 minor actions.
- Crouching to Standing is a minor action.

Minor Readying

The time taken to draw a weapon depends on its size and ease of use. The number of minor actions to ready or reload a weapon is listed in the description of each weapon.

- Changing Grips and Stances - weapons with versatile traits.
- Most weapons take one minor action to draw and
- Another minor action to reload, but some weapons are especially fast or slow.
- Minor Ready is the action needed to use a **Minor Loading Trait**.
    - Examples are Loading a Magazine, charging a grenade to be thrown.

Quick Scan

Reveals Anything or Anyone within line of sight or can be indirectly aware observable.

- Automatically Defeats Minor Stealth if adversary is within Line of Sight or Indirectly Knowable

Minor Stealth

Minor Stealth allows a character not to give away their position to a careless or busy adversary.

- Adversaries not using Quick Scan are not aware of this character.
- Move 2 spaces or 3m.

List of Significant Actions

Deliberate Stealth

Deliberate Stealth

- Automatically Defeats Quick Scan
- Roll stealth against Difficulty is equal 8 + Adversary's Recon + Adversary ChM + Adversary's Conditions to detect.
    - If there are Range, Concealment, Cover, terrain and other factors that make it harder to detect then this will lower the Difficulty to succeed in Deliberate Stealth.
    - A lot of Sensors, Characters performing Search/Scan can have a Superiority Modifier.
    - A lot of Visual Noise, Loud Threats, and other Stimuli may affect conditions.
- if the Character has an opportunity to displace without being detected the character can move 4 spaces or 6m.

Search/Scan

The character automatically Searches or scans for threats, or for something or someone.

- Automatically Defeats Minor stealth if the adversary is within Line of Sight or Indirectly Knowable

- Rolling Recon against Difficulty 8 + Adversary Stealth + Adversary ChM + Conditions.
  - If there are Range, Concealment, Cover, terrain and other factors that make it harder to detect then this will lower the Difficulty to succeed in Deliberate Stealth.
  - A lot of Sensors, Characters performing Search/Scan can have a Superiority Modifier.
  - A lot of Visual Noise, Loud Threats, and other Stimuli may affect conditions

Dash

The character moves 10 spaces or 15 meters in a roughly straight line.

- Roll athletics against Difficulty 8 + Conditions if there are Obstacles or bad footing.
  - The difficulty may be replaced by Adversaries or Threats DM if the character is Under Fire.
- Move 10 spaces or 15m in a straight line.
- Medium Load and Heavy Load can only move 7 spaces or 10m in a straight line.
- See also Movement Table, Running, and Minor Move

Significant Readying

Performing actions to reload a weapon, grab or take an item, activate a simple mechanism, etc...

- Example
  - sReady for short
  - Getting a Bulky weapon or equipment ready.
  - Getting a handful of ammo ready.
  - Getting equipment from an unready pouch.
  - Significant Loading. Regular Crossbows.

Reckless Action Option

A character can perform a task in a more dangerous or risky way that would be easier or more effective, but even if the character does succeed the character may be harmed.

- Reckless requires a Risk or Hazard, and braving that hazard makes it simpler or more effective for the character.
- A Reckless Action can be applied to any Significant Action.
- The PC/Adversary trades improved chances at greater risk.
- Gain a +2 to the Roll,
  - on a success, resolve the effects of the success.
  - but after calculating the effect with a -3.
    - If the Effect is negative, resolve the effects of failure.
  - on a failure, further increase or worsen the fail effect by 3.
- See Also Costly Success
  - The difference between Reckless Actions and Costly Success is that there is a chance a Reckless Action will have no Fail Effect unlike Costly Success.
  - Costly Success has a far greater Fail Effect.

Use Device or Skill

If a Skill or Device can be used within the span of a round then its possessed as a Significant Action. Example.

- Use Tactics Skill
- Use of Computers Instructing an Intellect Program to use an Expert Program or Agent Program.

Supporting Action Option

The objective of this action is to make the actions of other allies more effective. This is typically any Action that saves all other allies an Action to do.

- This action can be applied to a Significant Action.
- The PC supports another PC's action.
  - They sacrifice Success Effect
- The PC who succeeds adds their success Effect to the Ally PC or PCs roll.
- Example are
  - Suppression Fire. Shooting an opponent to keep them pinned down.
- The Difference with Coordinating Actions from Supporting Action is the Desired Outcome of Supporting Action is to Support or Aid.

Coordinating Action Option

The character's coordinate, making the Action more difficult

- Characters try to Coordinate to Performing Complementary actions. The Referee adds a 1-4 Difficulty to the coordination depending on the circumstance (number of character coordinating and their ability to coordinate).
  - Characters coordinating may gain Advantage, offset Disadvantage, or Gain Superiority.
  - Characters with Leadership and Tactics can reduce this Difficulty modifier.
- Tactics Skill may Offset this Difficulty modifier.
- The Difference with Supporting Actions is that both characters maintain their Objectives

- **Risks** - against an Unknown number of opponents Coordination actions may reveal an Adversary's superiority.

Defend

The character only chooses to defend. This is used strategically to occupy an adversary or buy time.

- Double a Defense Modifier
  - Choose either Cover, Concealment, or Skill.
- If skill is chosen to be doubled this also affects the DM of the roll, not just the calculation of Defense Modifier.
  - Example. A Character with Skill 1 and ChM 0, rolls on a 1 DM. If the Character has a Skill 2, the character rolls at a 4 DM. Skill matters in defense.
- On a Failure
  - if the PC is using a skill, reduce the Fail Effect by his skill
    - This is Doubled for the calculation of Defense Modifier if skill was chosen.
  - if the PC has Cover, Concealment, or some form of equivalent protection reduce the Fail Effect by the Conditions of the Cover, Concealment, or Protection.
    - PC Cover takes damage see Fail Effect and Damage to Protection.
- On a Success
  - Take no Damage.
- Note the maximum modifier of Cover and Concealment is 4. Defense does not improve these past 4.

Attack

Either Ranged or Melee Attacks.

- Attacks Roll the Weapon Skill + ChM + Other Modifiers against a Difficulty of 8 + relevant Conditions . This is Modified by the Adversary's:
  - Competence
  - Superiority
  - Conditions
- Keep note of Adversaries Conditions, which may be the same as the PC. Adversary Conditions are the Difficulties of the Adversary and will be the Mitigating Factors of the PC.
  - Range, Size, and Movement.
  - Example
    - Both Fighting Parties who engaged in Ranged Combat may have Range Size Move Modifier of 4.
    - On a Failed Roll these Conditions are Mitigating Factors
- Roll Weapon Skill + Characteristic Modifier + Skill against the Conditions set by the Referee
  - Defaults at 8. Plus the Char Modifier + Skill modifier of the most potent Adversary.
- On a Failure
  - if the PC is using a skill, reduce the Fail Effect by his skill.
  - if the PC has Cover, Concealment, or some form of equivalent protection reduce the Fail Effect by the Conditions of the Cover, Concealment, or Protection.
    - PC and Adversary's Cover takes damage see Fail Effect and Damage to Protection.
  - if Fail Effect is reduced to 0, PC suffers an Ordinary Failure.
  - otherwise the PC suffer a number of Attack Damage (minus protection) equal to the Fail Effect. See
- On a Success
  - if the weapon doesn't have any attack limitation (like crossbows and breech loaders) the PC deals attack damage.
    - Weapons with attack number limitations, roll a number of attack damage and choose the best one.
  - If the PC's Success effect is greater than 0, additional attacks deal damage.

Ranged Attack

An Action Option for Attacks

- Before Rolling declare Rate of Attacks and the use of Auto Fire.
- Difficulty Factors are:
  - Typically 8 Difficulty
  - Range, Size (Position), and Move
  - Cover and Concealment
  - Superiority
  - Aim
  - For Firearms - Rate of Attack used.
  - If under Fire, Exchange of Fire, or Awaiting and Ready Enemy.
- Example
  - 50m from the Adversary in Wilderness with Cover and Concealment the difficulties 8 plus 8 for 16.
  - Partisans with automatic bolt action rifles have an Aim bonus of 2, for a total of 3 when they are able to aim. If they take the entire round to aim they have a +5 DM from Aim. If they take

second round they have a +8 DM to shoot. If they catch the enemy unaware they hav a +8 DM against Difficulty 16.
    - The Partisan fails on a roll less than 8.
    - If the Partisan fails, and adversaries cannot shoot back then no consequence.
    - But if the Adversary can shoot back the Partisan gets hit if the margin of failure was greater than the Conditions of 8. rolling 1 is impossible so no counter attack.
    - So if the Partisan rolled 2 +8DM his Die Result is 10. His margin of failure is 6, but conditions being 8 the shot back doesnt hit.
  - If the Partisans is under fire, with Adversaries taking snap shots (one aim and fire) for a total Aim and Skill of 4, the Difficulty the Partisan has to beat is 20.
    - The partisan hits if they rolls a 12 for a total of 20.
    - Failing with a roll 11 +8DM, the partisan Dice result is a 19 his Fail Effect is 1. But with Conditions of 8 they does not get shot.
    - Failing with a Roll 3 +8 DM = 11
      - Dice Result of 11 - 20 Difficulty equals a Fail Effect of 9.
      - Fail Effect of 9 minus 8 Conditions (see Defense Modifier).
      - The partisan gets shot at a roll of 3 or 2 (if the Adversary only had an Aim and Skill of 4).
- See Dice Mechanics Diagram

Melee Attack

- Difficulty Factors are:
  - Typically 8 Difficulty
  - Adversary Competence
  - Cover from Shields
  - Superiority
- Example
  - A Bandit Adversary with a Skill DM 2 and none from Characteristics with Axe and Knife has a 10 Difficulty or as a PC has a defense modifier of .
  - A Levy Footman with long knife and Medium shield (1 Cover) with Skill DM of 1 and Str of 1 as an Adversary has a 11 Difficulty or as a PC has Defense Modifier of 3.
  - The Levy as a PC making a melee attack against the bandit must roll better than 10, but will get injured if they fails by 3 or more (from Shield and Skill).
  - The Bandit has to strike the
- See Shields and Cover,

Attacks and Superiority

- Apply the context that multiple attackers have greater combined number of RoF/RoA.
- When an numbers diminishing or get incapacitated Superiority diminishes.

Attack Options

Moving Attack Option

Moving Attack option is when the character incorporates moving while attacking. Moving attacks only cover a distance equal to a Movement.

- See Combined Difficulties when incorporating Attack and Movement.
- Examples
  - *Fire and Displace* - Ranged Combat and Stealth - if the weapon is practically silent.
  - *Moving to Cover while Attack* - Ranged Combat and Athletics.
  - ***Leaping Attack*** - Athletics, Melee Combat, Reckless Attack Option - attacking from higher ground and with the momentum of falling.
  - *Bull Rush* - Athletics, Melee Combat, Grappling Attack option - aiming to knock down

Targeted Attack Option.

making targeted attacks increases the Difficulty, in return for more lethal or specific effect - like disarming, or immobilizing .

- See Targeted Damage
- use 2d6 in the Targeted Attack Location table to determine random Hit Location when necessary.

Targeted Attack Location table

2d6	Odds	Diff	Location
2	2.78%	3	Vitals, Neck
3	5.56%	2	Feet*
4	8.33%	2	Secondary Leg
5	11.11%	1	Primary Leg
6	13.89%	1	Groin
7	16.67%	0	Core or Center of Mass.
8	13.89%	1	Chest
9	11.11%	1	Primary Arm
10	8.33%	2	Secondary Arm
11	5.56%	2	Hands*
12	2.78%	3	Head

* roll 1d6 1-3 Primary, 4-6 Secondary

Ballistic Correction.

- Characters who continue to shoot at the target can gain DM to hit.
  - If the effects of the shots can be seen by the shooter then they can keep their Aim bonus from the last round and increase it by 1.
  - A spotter who can call out coordinates can grant a 2 Conditions as well.
  - **Tracer Rounds**. can grant up to 1 Aim bonus, up to 3 aim bonus on fully automatic (being able to see the stream of fire). These also aid adversaries in finding and shooting their source.

Grappling

A character can attempt to wrestle or grab another person instead of hitting them. The attacker must move to Personal range and beat his target in an opposed Natural Weapons skill check. If they wins, they may do any one of the following:

- *Step 1: Grappling Action*
  - Grappling is a Melee Combat Action - DM from Str or Dex of the Adversary modifies the Difficulty.
  - Unskilled Characters. PC Without unarmed combat suffer -3 for being unskilled, unskilled adversaries reduce the DC by 3.
  - factor Close quarter Weapons like knives, stun gloves. This factors the difficulty by 1 if someone has this over the other party.
  - Pistols - firearms that can be employed in close quarters increase the difficulty by 2.
  - Cumbersome objects like shields, rifles, spears, etc.. can affect grappling Difficulty.
- *Step 2: Grappling Roll,*
  - just like a regular Melee Attack Roll.
    - Characters can attempt a specific Grappling Maneuver.
      - Targeted Limb Difficulty +1
      - Targeted Extremity Difficulty +2
      - Throw Difficulty +3
      - Sweep Difficulty +2
- *Step 3: Grappling outcome*
  - On a Success the character gets an advantageous position (example - shooting to the flank or rear of the adversary), the Character gains an advantage on the next round.
  - On a Fail the character gets a disadvantage on the next round.
  - The Success or Fail Effect can be used to determine Grappling Effect. The Ref for the Adversary or the PC gets to choose a combination of Target and Damage or Bind.
  - With enough Success: Damage and Bind can be applied.
  - If the Damage is intended to Knockback or Knockdown.
    - If the Damage = 1/2 serious wound its a Knockback even if it doesn't penetrate armor.
    - If the Damage ≥ serious wound its a Knockback even if it doesn't penetrate armor.
  - Multiple Targets can be affected with enough Fail or Success Effect.
  - Example a Success Effect of 6 can be used to deal
    - 1d3 damage (2) against an hand (3)
    - 2d3 damage (4) against an arm (2)
    - Bind (2) both arms (4)

Grappling Maneuver Table

Effect	Specific	Effect/ Difficulty	Notes.
Target	Limb	2/1	Target a limb with Bind or Damage
Target	extremity	3/2	Target a hand/foot with bind or damage.
Damage	Body, limb or extremity	2/1	deal 1d3 bl to the target. this is twisting of limbs and extremities and armor does not affect this damage. No target assumes a body Throw. Armor does not protect against this attack.
Bind	Target	2/1	render the target unusable or bound while grappled.
Throws	Body	Special/3	Throw an adversary with Effect equal to 1/6

		(round up) their HP. Target is lying down. Takes 2d3 bl damage.
Sweeps Legs	Special/2	Throw an adversary with Effect/Diff equal to 1/2 (round up) their Dex. Target is Prone.

Grappling is a tactic used when trying to negate an adversary's more lethal reach or range, trying to capture, or trying to surprise an adversary.

Thrown Weapons

- Thrown weapons have a thrown property range equal to the Strength in Meters with modifiers. See Thrown Trait
- The value after the thrown property is how far the object can be thrown accurately level.
- Thrown Value x3 is the distance a weapon or object can be Lobbed - thrown at a height to allow it to travel farther before it hits the ground.
- Example
  - Thrown 0.5, a character with Strength 7 can only throw this weapon 3m straight, or 10m lobbed. .
  - Thrown 1.5 allows a Str 7 character to throw 10m straight or 31m lobbed.

Plan

Plan action allows the character to react to a Predicted or Predetermined Trigger Condition. Gain and advantage if the prediction is correct, or a disadvantage if not.

- **Step 1:** The PC declares the scope of their Plan.
  - Example: “I will try to anticipate the enemy’s Actions”
- **Step 2:** The Referee determines what the Adversaries Actions will be
  - Examples: the Enemy is going to Flank the PC’s right.
- **Step 3:** The PC declares the Trigger Condition, and his Action AND what they does when the Trigger Condition does not Occur.
  - Example: If Enemy tries to circle to my Left (Trigger Condition), I attack them (Action with Advantage). Otherwise I Defend (Action with Disadvantage).
- **Step 4:** Resolve if the Prediction is correct or not to determine if the roll will be made with an Advantage or Disadvantage.
- **Another way to Resolve**.If the Player happens to divulge the Prediction without the Referee having a chance to consider the Referee can simply modify the Difficulty depending on how likely or unlikely is the prediction.
  - Depending the the Likelihood the Difficulty is modified positively or negatively by 1-4. .
  - if the prediction is not specific enough to be exploitable then it is resolved like a regular action.
- A Common use of Plan is to Plan in case an Adversary is within Line of Sight (Trigger Condition) to Attack, but Otherwise Aim (minor action) or Move.

Trigger Condition

Trigger Conditions are used in Plan Action Options. The Character declares the conditions that will trigger their Action and what action they will perform otherwise.

- Example Trigger Conditions.
  - Targets or Adversary present themselves in the Area the PC is Covering and Aiming in.
  - To act Immediately after a pause of Gunfire or movement.

List of Full Actions

Full Skill Action

Full Skill Action are actions that take more than a significant action to perform and typically extend past the initial round.

- First Aid is an example of a Full Skill Action, Trauma Kits are Crucial

Running

The character runs in a Straight Line, dodging or hurdling over obstacles. the character may have to roll against attacks made against them.

- Roll Athletics + End or Dex
- Running grants a Move Difficulty Adversary Condition which becomes a Defense Modifier.
- Running on Light Load is 15 spaces or 22.5m
- Running with Medium and Heavy Load is only 10 spaces or 15m
- See Minor Move, Dash, and Movement table

Extended Actions

- Some skill checks will take longer than a single combat round to complete.

- Make a Timing roll for the task and then work out how many six second combat rounds it will take to complete. Example is First aid takes 30 rounds.
- A character engaging in an extended action cannot do anything else and must either Abort their action to defend themselves or endure the hazard or focus on the task. See Full Action

Special Considerations

The following are special considerations in Personal Combat.

Tactics Skill

- Tactics is rolled at the beginning of Combat. It grants an amount of Advantage or Disadvantage Rolls the PC will gain.
  - PC must declare how risky the tactics they will attempt. 6, 8, 10, 12
  - Risk determines maximum Failure and Success Effect.
    - Maximum Effects of 6 Difficulty is 1.
    - Maximum Effects of 8 Difficulty is 2.
    - Maximum Effects of 10 Difficulty is 4.
    - Maximum Effects of 12 Difficulty is 8.
- When the Tactic Effects are Exhausted (all the Advantage or Disadvantage rolls are used up), the Character can try again
  - if the conditions of the battle has changed a lot according to the referee or
  - if the Character's number of attempts is less than his skill level in Tactics+1. A character with skill1 in Tactics can try 2x in a battle.

Two-Weapons

- Two weapons allows for a Higher rate of Attack, opportunity cost is that of a shield's cover.

Shields

- **Small Shield**.
  - Grants a cover Conditions +1 to characters skilled in melee combat. Characters with combat skill at 0, benefit from a small shield.
- **Medium Shield**.
  - Grants cover Conditions +1,
  - Grants cover Conditions +2 when crouching low.
- **Large Shield**.
  - Granting cover Conditions +2
  - Granting cover Conditions +4 when crouching low.

OUTCOMES

Damage

This is the most common outcome in combat actions.

- Damage can come from
  - Weapons
  - Environmental Hazards
- Successful Attacks Deal Damage to Adversaries. Costly Failures or Fail Effects mean Attacking Adversaries or Hazards Deal Damage to the Player Character.
- Success Effect determines the Number of Attacks the Adversary takes. See Damage Rolls.
- and Fail Effect determines the Amount of Attacks the PC takes. See Damage Rolls
- Example.
  - Automatic Rifle with 3d6-2 of Damage has a Success effect of 2. If the Character made as many attacks Possible, and the Rate of Attack was 4, then the Automatic Rifle deals 3d6-2 damage 3 times,
    - First on the Success.
    - Second and Third successful Attacks from the Success Effect of 2.
    - If the Difficulty was 12, and the Character has a Dice Result of 14, then the success effect would be 2.
- Weapons with a Limited Rate of Attack, and the Success or Fail Effect exceeds the Rate of Attacks, results in as many Damage Rolls BUT only choose the highest Damage rolls equal to the Rate of Attacks.
- Example
  - A Fail Effect of 3, after accounting for All the Conditions and Mitigating Factors, the PC will suffer 3 Attacks if the Weapon had a rate of attack of 3 or greater. But if the Weapon had only a rate of attack of 2, then 3 rolls would be

made and the Two highest Rolls will be chosen as damage.

Targeted Damage

- Targeting the Limbs, Extremities, or Vitals have special effects.
- Limbs Taking ≥ 1/3 HP are crippled
- Extremities Taking ≥ 1/10 HP are crippled
- Heads Taking ≥ 1/10 HP are stunned
- Vitals Taking ≥ 1/10 HP causes bleeding.
- See Targeted Attack Location Table and Targeted Attack Option

Damage Scale.

- Deka for tens, Heka for hundreds, Kilo for Thousands. Typically these are used for weapons designed to take out Buildings, Vehicles, and Ships.
  - A vehicle, character, armor, or object of a Default Damage scale can ignore damage of a lesser scale. Take note of the default scale.
  - D,H,K before the dice is used to denote the damage scale. Example
    - D 3d3 = 3d3 x10
    - H 2d6-2 = 2d6-2 x 100
    - K 5d6-4 = 5d6-4 x 1000

Armor

- Armor reduces damage by the value of the armor.
  - Every attack has their Damage Reduced by Armor.
- A hit with Effect 6+ always inflicts at least one point of damage, regardless of the target's armor.
- Example:
  - A character suffers a Fail Effect of 3, after all conditions and Mitigating Factors have been factored, and there are multiple adversaries allowing for 3 attacks with a Broadsword of 3d3-2.
  - The character has a Flak Jacket of Armor 4. The Adversaries will roll 3d3-6 three times.

Armor Scale

- Deka for tens, Heka for hundreds, Kilo for Thousands. Typically these are used for weapons designed to take out Buildings, Vehicles, and Ships.
- Example
  - DAR 2 = 20 AR
  - HAR 9 = 900 AR
  - KAR 3 = 3000 AR

Hit Points (HP)

Characters, Objects, Vehicles, Infrastructure have hit points or simply called Hits or HP.

- Hit points are a Threshold - a value used to compare damage against to determine how badly injured the Character, Object and Vehicle Can be.
- Characters have HP equal to the sum of their three (3) physical characteristics.
- Take note of HP and thresholds. For characters the thresholds are:
  - Wounded. taking any amount of damage.
  - Wound Penalties. divide the Damage taken by 3, round down. This is the penalty to all mental and physical rolls.
  - Seriously Wounded. taking an amount of damage greater than or equal the character's best physical characteristics.
  - Critically Wounded. taking an amount of damage equal or greater than 2 physical characteristics.
  - Dying. Taking damage greater than HP.
- Notes.
  - Recommend using Tally Marks in groups of 3.

Hit Point Scale

- Deka for tens, Heka for hundreds, Kilo for Thousands. Typically these are used for Buildings, Vehicles, and Ships.
  - A vehicle, character, armor, or object of a Default Damage scale can ignore damage of a lesser scale. Take note of the default scale.
  - D,H,K before the HP value is used to denote the HP scale. Examples
    - 20 DHP = 200HP
    - 3 HHP = 300HP
    - 67 KHP = 67000HP

Damage Rolls

The Success or Fail Effect determines how many Damage rolls are made.

- Damage Rolls are individual sources of damage (like strikes or blows or effects) made by Adversaries, or risk and harm in the Environment, and Strikes and Blows made by the PC.
  - A fail effect of -1 results in a number of Damage roll against the PC equal to the fail effect. Up to the sum of the Rate of Attacks of the Adversaries or Applicable hazards.
    A fail effect of -2 results in 2 damage rolls.
    - See Mitigating Factors

- The Adversary weapon or the Hazard's damage determines the damage from a Fail Effect.
- Apply Protection or Armor to each Damage Roll. Example.
  - a Success effect of 1 results in two (2) 3d3-2 broadsword damage rolls.
    - Against an Armor of 3 Roll two (2) sets of 3d3-5.
  - A Success effect of 3 results in three (3) 3d6 automatic rifles with a RoA 3 has damage rolls, it would have been 4 if the rifle had a higher rate of fire.
    - Against an Armor of 5 Roll three (3) sets of 3d6-5.

- An Attack Action can result in Multiple Strikes. See Rate of Attack depending on the Effect.

Changes of Conditions

After every round take note of the changes of the conditions.

- Battlefield Conditions
  - Ranges, Cover, Visibility/Concealment, Terrain, Positions, etc…
  - Superiority
- Character Conditions
  - See Physical and Mental Conditions.

CHARACTER CONDITIONS

On the Brink

Characters who lose their turn due to the Player Character Conditions are considered "On the Brink". This rule is in place to allow players whose characters are unable to act to shape the story.

- If the Bound, Immobile, or Helpless PCs cannot take an action to affect their condition (Example: rolling a skill to escape or oppose a grapple) then they can
  - Grant an Advantage to a PC of their choice, but the other Player gets to add to what happens on their turn.
- The Player of a Dying PC, during their turn must state a course of action they will do if they survive, in light of their Near Death condition.
  - They Gains Advantage for future Rolls, relevant to what they intend to do when they survive.
  - This transfers to any Character the Player chooses, even future characters.

Mental Conditions

These include conditions the ability to be aware, think, perceive, communicate, and speak.

Unaware

Characters which are unaware have a Disadvantage in defending themselves, or Attackers have an Advantage attacking them. Blindsided or Flatfooted can mean Unaware.

Dazed

Strong emotions, Context overload, Confusion is able to Daze Targets.

- The character can only Move or Defend.

Stunned

Strong emotions, Fear, Shock, Confusion is able to Daze Targets.

- Unable to take any most Physical and Mental Actions, unable to fully defend themselves. Reduce the Difficulty to attack the Stunned Target by 2.
  - PCs who are Stunned have the Difficulty of all their Actions increased by 2.
  - The Difficulty to attack Adversaries who are Stunned is reduced by 2..

Panicked

Overwhelming Fear causes the character to take defensive actions

- PCs who are Panicked are Disadvantaged when they make Attacks.
- PCs who are Panicked may take only either Defensive or Move Actions

Routed

Overwhelming Fear causes the character to retreat and flee.

- The only actions Routed PCs may take are Defend, Retreat, or Pull Back.

Physical Conditions

Light Load

- Characters can use Str or the Average of Endurance and Strength. Whichever is higher to determine their Load.
- Characters can carry up to twice their Str in Kilograms without any Penalty.

Medium Load

- Characters can use Str or the Average of Endurance and Strength. Whichever is higher to determine their Load.
- Characters can carry up to four times their Str in Kilograms.
- While under Medium Load the Character is considered performing Moderate Activity.
- Medium Load Causes the Character to suffer
  - -1 DM Load Penalty to Physical Rolls. This penalty applies in Low to Zero G, as the Load affects the Inertia the character has to manage.
  - -75% movement.

Heavy Load

- Characters can use Str or the Average of Endurance and Strength. Whichever is higher to determine their Load.
- Characters can carry up to six times their Str in Kilograms.
- While under Heavy Load the Character is considered performing Strenuous Activity.
- Heavy Load Causes the Character to suffer
  - -2 DM Load Penalty to Physical Rolls. This penalty applies in Low to Zero G, as the Load affects the Inertia the character has to manage.
  - -75% movement.

Maximum Load

- Characters can use Str or the Average of Endurance and Strength. Whichever is higher to determine their Load.
- Characters can carry up to twelve times their Str in Kilograms.
- While under Heavy Load the Character is considered performing Intense Activity.
- The character cannot perform any other Action except but hold up the Maximum Load.
- A Full Action allows the Character to Carry the Maximum Load over 1 space (1.5m space)
- See Also Push/Drag

Load	KG	DM	Move
Light	Str* x2		
Medium	Str* x4	-1	x0.75
Heavy	Str* x6	-2	x0.75
Maximum	Str* x12	No other Action	1.5m/rnd
Push Drag	Str* x30	This is an Action	x0.5

*Str or the Average of Str and End
*Overloaded

Load Table

STR	Light	Medium	Heavy	Max	Push/ Drag
2	4 kg	8 kg	12 kg	24 kg	60 kg
3	6 kg	12 kg	18 kg	36 kg	90 kg
4	8 kg	16 kg	24 kg	48 kg	120 kg
5	10 kg	20 kg	30 kg	60 kg	150 kg
6	12 kg	24 kg	36 kg	72 kg	180 kg
7	14 kg	28 kg	42 kg	84 kg	210 kg
8	16 kg	32 kg	48 kg	96 kg	240 kg
9	18 kg	36 kg	54 kg	108 kg	270 kg
10	20 kg	40 kg	60 kg	120 kg	300 kg
11	22 kg	44 kg	66 kg	132 kg	330 kg
12	24 kg	48 kg	72 kg	144 kg	360 kg
13	26 kg	52 kg	78 kg	156 kg	390 kg
14	28 kg	56 kg	84 kg	168 kg	420 kg
15	30 kg	60 kg	90 kg	180 kg	450 kg
16	32 kg	64 kg	96 kg	192 kg	480 kg

Physical Conditions

Physical conditions affect the characters ability to move, recover, work, and coordinate.

Exhaustion

- Exceeding the Capacity to perform Activities (see Light Activity) the Character must Roll End Difficulty 6 plus the hours exceeded.
  - Example. At a Light Activity Capacity of 14 hours, performing Work in the day that is equal to 15 triggers an End roll Difficulty 5.
  - Failure - take 1 damage plus 1 damage per fail effect of 3.
  - Success - no damage. No effect
  - Athletics Endurance skill only helps with Physical rolls.
- Damage from Exhaustion is noted as Exh Dmg.
- Like Wound Penalties every 3 Exhaustion Damage gives -1 penalty to All rolls.

- Character's heal Exhaustion damage like regular injuries.
- Stims allows to Ignore Penalties of Exhaustion for a number of hours.
  - Stims 6 hours level 1 removes only up to 1 level of penalties. 50cr
  - Every Additional Stim per 24 hour period deals exhaustion damage equal to its Potency, which its level.
  - Stims 6 hours level 2 is 150cr.

Resting
- Resting is performing very minor activity, far easier than Light Activity.
- This is typically eating, relaxing, chatting, and watching.

Light Activity
- Examples of Light Activity
  - Marching or Walking 5kph or 1.38m/s
  - Characters under Under Light Load are considered performing Light Activity.
  - Being under uncomfortable heat and cold counts as such
- Characters can perform End x 2 hours of Light Activity per Day.

Moderate Activity
- Examples of Moderate Activity
  - Characters under Medium Load
  - Character on a Light Jog 7kph or 2m/s
  - Picking and Transfering.
  - Physical labor, Carpentry, Machining, Assembly
  - Being under difficult heat and cold counts as such,
- 1 hour of moderate activity equals 2 hours of Light activity.

Strenuous Activity
- Examples of Strenuous Activity
  - Characters under Heavy Load.
  - Characters in a light Run of 10kph or 3m/s
  - Brute force labor Shoveling, Digging, hauling and moving heavy equipment and controls.
  - Being under extreme heat and cold counts as such
- 1 hour of moderate activity equals 3 hours of Light activity.

Intense Activity
- Example of Intense Activity
  - Characters under Maximum Load.
  - Characters in a Running of 15kph or 4m/s
  - Feats of maximum strength and endurance.
  - Being under inhospitable heat and cold counts as such
- 1 hour of moderate activity equals 3 hours of Light activity.

Knocked Back
The character is moved by one or more space, depending on the Referee. Typically bludgeoning damage relative to their Serious Wound.
- If something can be knocked back it would be a number of spaces equal to the bludgeoning damage (ignore armor from this calculation) it took is ½ its Serious Wound.

Knocked Down
The character is knocked down to prone or lying down, depending on their orientation.
- If something can be knocked down it would be because it took bludgeoning damage (ignore armor from this calculation) ≥ ½ its Serious Wound and it had bad footing or an imbalancing factor.

Grappled
- Grappled Characters are considered physically engaged in unarmed combat with another character.
- Grappled Conditions are like bound or crippled, that it can Affect Limbs and Extremities.
- See Grappling Action
- helpless and Immobile

Bound
- Characters who are Bound are typically unable to use their arms, hands, legs, and feet. Unless someone frees them from the conditions.
- Bound Characters are similar to crippled in that they cannot use the Limbs or Extremities that are bound.
- Bound characters typically are considered helpless and Immobile unless they have some limbs or extremities unbound that allow them to move or defend themselves.

- Even if they have difficulty moving or defending themselves they are considered having disadvantage to their roll.
- See Also helpless, crippled, Immobile, On the Brink,

Stat Damage

- Characteristic Damage. Damage to Characteristics are logged and noted, and the Stat is compared to the Damage.
- The stat functions equal to the difference of the Stat less the stat damage.
- Medical Treatment, First Aid, Surgery, can be used to reduce Stat Damage.

Blind

- Characters suffering the Blind conditions cannot Directly See,
- must use non-visual means to Indirectly Know where a threat or target may be, and
- they suffer the conditions of Invisible Threat If they cannot determine the source of attacks and defend appropriately.
- See also Indirectly Know, Directly See, Invisible Threat, Attacking while Invisible, Concealment

Crippled

- A condition to be used to describe a limb (leg, arm) or extremity (hand or foot) being rendered unusable or inoperable.
- A character with a crippled leg or foot cannot move without a crutch or another character acting as one.
  - With a crippled leg or foot, the character can only drag themselves over the ground at 1/4 their move with their arms.
  - If the character's hand is crippled but still can drag themself by his elbows and forearms they can move 1m per round as a full action.
  - If the character's crippled to the extent they cannot move they are considered Immobile and possibly helpless.

Immobile

- A character is immoble if they cannot move from his position or space.
- if they are being Grappled, they can only take their turn to oppose the grapple. see Grappling
- See Also helpless, crippled,

Helpless

- A character is considered helpless if they cannot physically defend or move themselves from harm's way.
- PCs who are helpless still roll on their turn,
  - they roll the Skill or Dice Modifiers relevant in Surviving the Attack - typically Endurance plus the conditions that favor or hamper their ability to take the injury.
- See Also Immobile, crippled, Full Action, On the Brink

Unconscious

- Synonymous with Knocked Out, our Out.
- The character is not conscious and cannot perform any mental or physical actions.
- the character is considered helpless and Immobile,
- Characters who are Critically Wounded are automatically considered Unconscious and Knocked out.

Situational Condition

situational conditions are unlike mental or physical conditions. This depends on the when and where that affects the character.

Under Fire

When the adversary has the Initiative and Covering an Area with Fire, its harder to act or return fire.

- When characters are Attacking or Performing actions against an enemy that can Directly See them or the action will cause the Enemy to Directly See them the Adversary's Skill and Aim Modifier (if applicable) adds to the Difficulty.
- There are many actions that avoid this penalty,
  - staying under cover or
  - moving quickly from cover to cover
  - Supporting Actions - this Causes the Adversaries to allocate their attention to the Supporting
  - Coordinating Actions -this Causes the Adversaries to either quickly decide how to allocate their attention or fail to use it effectively.
- Synonymous with Taking Fire

Concealed or Hidden

the character is undetected and characters are unaware of this character.

Flanked

The character is between or facing multiple threats, and has to divide attention and resources. They suffer Superiority Conditions modifiers.

Healing and Recovery

Natural Healing

- Healing is Reducing Damage at a rate over a unit of time.

Full Rest

- Full rest is 8 hours of sleep, and No Light, Moderate, Intense, Strenuous, Stress or Active Physical or mental Activity.
- If the Difficulty is greater than 4 it has some stress. Typically these are tasks the Referee doesnt need the PC to roll for.
- Full rest is interrupted when Light and Strenuous Activity in taken.
  - If Interrupted, Full Rest can begin again on the next day.
- Those in bed rest are considered to be in Full rest.
- Maintaining a Resting heart Rate (which is 40-80 beats per minute).
  - Challenging Clerical or Pape work or Knowledge Work can be strenuous.

Medical Treatment

Natural healing is not the only way to recover damage. Characters can also seek out medical treatment.

First Aid

- Applying first aid reduces an amount of damage equal to the Effect of the Medic check, at minimum 1.
- It takes a minor action to get the materials from a ready First Aid Kit and Significant Action to perform first aid.
- Difficulty 8 to perform first Aid on another character.
- Difficulty 10 to perform First Aid on oneself.
-
- It takes 3 mins or 30 rounds to perform First aid. A character can shorten by a number of rounds by their success effect roll every round.
  - They lose such progress by their Fail Effect. Their fail effect determines the number of additional rounds they have to perform first aid.
  - See Trauma Kits which gives DM bonus for First Aid.
- Character's who perform First aid, will be unable to do anything else for a number of rounds depending on their Success or Fail Effect plus 3.
  - On a regular Success Effect of 0, 3 additional rounds for first aid. Reduce the number of rounds needed by 1 for every Success Effect over 0.
  - This assumes the Injured character can hold their own bandages and injuries.
- First Aid is Important for the following:
  - to Stop Bleeding
  - to Stop Dying Rolls temporarily for a number of Hours equal to the Dying Character's End.

Surgery

- A character that is seriously wounded (after first aid has been applied) requires surgery.
- Difficulty 8, 30 minutes Task Duration.
  - Tools, Assistance, etc modify the roll.
- Surgery reduces damage equal to twice the success effect.
- Fail effect.
  - Reduce the Fail Effect by the Medic Skill. Any remaining Fail Effect is the multiplier to the time of Surgery.
  - Fail Effect Exceeding the Half Character's End damages the Character by double the Fail Effect.
- Surgery requires a hospital or sickbay.

Medical Care

Medical care removes damage at an Increase Rated depending on the resources, tech level, man power and tools.

- Typically there are no Rolls on Medical Care, unless its Intensive Care.
- Character with Medic 0 can care for 8 light care patients. Every level of Medic above 0 doubles this.
  - Medic 1 = 16 light care patients
  - Medic 2 = 32 light care patients
  - Medic 3 = 64 light care patients
  - Medic 4 = 128 light care patients
  - Tools, Technology, Robots, and other Resources multiply this effectivity.

- Higher Level Medic at 2 and above can double the care capacity of the medics they are supervising.
  - This assumes the medic is checking and examining the decision of the medical staff for technical soundness.
  - Leadership and Admin will be necessary for
- Damage causing injury is reduced at double the rate at 2 less with Natural Healing and Full rest.
- Characters with Medical Care can perform Light Activity but this affects their rate of healing.
  - Wounded characters reduce damage at 1 per day of Light Activity.
  - Seriously Wounded/Injured reduces damage at 1 damage every 3 days with Light Activity and Moderate Care.
  - Critically Wounded/Injured reduces damage at 1 damage every 7 days with Light Activity and Serious Care.

Light Care

- Characters that need little monitoring, just making sure the patient is resting, taking their medicine, drinking, and eating properly.
- These injured or sick can take most care of themselves.
- While the character has Damage < 1/3 HP they only require light care.
- A character with medic 0 can monitor 32 light care patients in a day

Moderate Care

- The caregiver must help in some minor therapy, application of medicine, and cleaning, other than basic care.
- While the character has Damage > 1/3 HP they require moderate care.
- 1 moderate care patient = 2 light care patients.

Serious Care

- These sick and injured characters need to be fed, changed, exercised and moved.
- While the character has Damage > 2/3 HP they require serious care.
- 1 serious care patient = 4 light care patients.

Intensive Care

- These sick and injured characters are similar to the Serious care but there is a need for a Medic 2 character or an AI sickbay to supervise and instruct Otherwise:
  - The character has to roll Difficulty 8.
  - Failure is when the character has to make a Dying Roll.
  - Success is that Character is considered to have benefited from Medical Care and heals (reduces) 2 damage (or more depending on Tools and Technology).
- While the character has Damage > HP they require intensive care.
- Characters with the skill of Medic 2 or greater do not need to roll when they have tool, help, and resources.
- A character who is Dying but receiving Intensive Care is typically immobile and deducts 1 point of damage every 3 days.
- 1 Intensive care patient = 8 light care patients.

Healing and Mental Characteristics

Other than Psionic Strength, characters may also suffer damage to their Intelligence or even their Education.

- Characters can recover from Characteristic Damage once the character has fully healed.
- The character recovers reduces 1 stat damage per 3 days of Physical or Psychological Therapy. The character can have therapy for 1 stat at a time.
- Note: use of Psionics fall under Exhaustion Rules, each Psionic Point is equal to 1 hour of moderate activity.

Injury

Injuries are conditions that the character has when they take any sort of damage - wounds from weapons, attacks, or hazards, or exhaustion from physical stress.

Hitpoints

PCs have Hit Points equal to the sum of their Str, Dex, and End

Wounded

Synonymous with Injured

- Taken any damage below 1/3 HP
  - Character can be considered Wounded from damage from Exhaustion.
- Healing and Recovery while Wounded
  - The character performing Moderate Activity cannot heal.

- Moderate Activity with Light Care has 1 less damage per 3 days.
- Light Activity heals 1 less damage per 2 days.
- Light Activity with Light Care OR Full Rest by itself is 1 less damage per day.
- Full Rest with Light Care is 2 less damage per day.

- Lost Progress, performing more strenuous activity loses the days progress, and the character has to start their healing progress the next day.

Wound Penalties

Every 3 points of damage. Penalties to all Die rolls.

- Morphine, Combat Drugs, or Pain Management Drugs can reduce Wound Penalties.
- Pain Management drugs Level reduces an amount of Wound penalties equal to the level. The drugs have a Duration.
  - Pain management Drugs can be addicting, character must roll End Difficulty 8 to resist the craving after taking the drug for more than 2-3x in a period of a month. Any HABIT or recurring use of it triggers the Roll.
  - Failure Effect is the number of Days the Character suffers a -1 to all Rolls that day.
- See Medical Consumables

Seriously Wounded

Taking an amount of damage equal or greater than a third of HP rounded down. *Synonymous with Seriously Injured*.

- Damage ≥ 1/3 of HP (rounded down), but <2/3 of HP.
- Example:
  - Character with 21HP, takes 4 attacks of 2, 4, 3, 1 for a total of 10 damage. 10 > 7, the character is seriously wounded.
- Seriously Wounded characters cannot move that fast, their movement speed is at 3/4.
  - Move 6m /4 spaces ⇒4.5 m / 3 spaces.
  - Move 4.5m / 3 spaces ⇒3 m / 2 spaces.
  - Seriously Wounded PCs cannot Dash
- Characters can be considered Seriously Wounded from damage from Exhaustion.
- Healing and Recovery while Wounded
  - The character performing Moderate Activity cannot heal. Note the Pain and Wound Penalties persisting while being Seriously Wounded.
  - Moderate Activity with Moderate Care does not heal the character but Psychologically better because of the pain management.
  - Light Activity heals 1 less damage per 3 days.
  - Light Activity with Moderate Care OR Full Rest by itself is 1 less damage per 2 days
  - Full Rest with Moderate Care is 1 less damage per day.
- Lost Progress, performing more strenuous activity loses the days progress, and the character has to start their healing progress the next day.

Special Conditions

- See Bleeding, .
- See Wound Penalties.

Bleeding

Taking an amount of damage or one damage roll equal or greater than being seriously wounded from one damage roll triggers a Bleeding Condition.

- 1 damage roll that is ≥ 1/3 HP rounded down.

Example

- A character with 21HP, takes 2 damage rolls of 3d6-2. The results were 4 and 9. The 9 damage causes bleeding.
- The character will continue to take 1 point of damage every minute,
  - starting 1 minute after the damage was taken.
  - The character can feel getting weaker and if the bleeding can be seen other characters will be able to tell.
- The Application of First Aid cancels this condition.

Critically Wounded

Critically Wounded Characters are typically knocked out or Unconscious.

- Damage is ≥ 2/3 of HP (rounded down).
- A character who is Out is helpless.
- Example:
  - Character with 21HP, with 10 pts of damage takes more damage, damage rolls resulting in 2, and 4. Exceeding 14, the character goes out.
- Healing and Recovery while Wounded
  - Being Critically Wounded renders the character Immobile and typically can only take full rest. With Full rest the character recovers -1 damage per 7 days.
  - Full Rest with Serious Care is 1 less damage per 3 days.

- Lost Progress, performing more strenuous activity loses the days progress, and the character has to start their healing progress the next day.

Special Conditions.

- Adrenaline Shots or other Combat Stimulants can awaken a character in this condition. Violent Actions causes the character to have the Bleeding Condition.
- Powered Suits with Remote Controls or Intellect Program will "Puppet" the character to bring them into a cover and safety. If the Powered Suit has Expert: Medic and Intellect programs it can perform First Aid.

Dying

- Damage is > Hitpoints.
  - Every Round PC makes an End roll against Difficulty 4 (note the Wound Penalties apply to this roll).
  - On a failed roll the character is dead.
  - On a success the PC is still alive.
  - First Aid stabilizes the character equal to their End in hours, where the character needs Medical Treatment.
  - The Dying character will continue to need First Aid until the character can get Intensive Care
- Dying Characters are Unconscious and Immobile
- Dying Characters require Intensive Care.
- Characters with Medic The character has to roll Intensive Care at Medic Difficulty 8.
  - Failure is when the character has to make a Dying Roll.
  - Success is that Character is considered to have benefited from Medical Care and heals (reduces) 1 damage (or more depending on Tools and Technology) every 3 days of Intensive Care.
  - Characters with medic 2, tools, help, and other resources do not need to make daily Intensive Care rolls.
- Healing and Recovery while Wounded
  - Dying renders the character Immobile and typically can only take full rest. With Full rest the character recovers -1 damage per 7 days.
  - Full Rest with Serious Care is 1 less damage per 3 days.
- Lost Progress, performing more strenuous activity loses the days progress, and the character has to start their healing progress the next day.

Massive Damage

- Characters who take more than their HP in total damage from an attack must roll End difficulty 8 with wound penalties.
  - Surviving could mean all the damage was concentrated on one area that would allow the character to survive - this would leave the character crippled if they managed to survive.
  - Characters who have damage > their HP need medical treatment until they are back to
-

Natural Healing and Medical Care Table

	Moderate Activity	Moderate Activity with Adequate Care	Light Activity	Light Activity with Adequate Care	Full Rest	Full Rest with Adequate Care
Wounded (Dmg < 1/3 HP)	No Healing	-1 dmg/ 3 days	-1 dmg/ 2 days	-1 dmg/day	-1 dmg/ days	-2 dmg/ day
Seriously Wounded (Dmg < 2/3 HP)	No Healing, note the Wound Penalties and pain	No Healing	-1 dmg/ 3 days	-1 dmg/ 2 days	-1 dmg/ 2 days	-1 dmg/ day
Critically Wounded (Dmg < 2/3 HP)	Immobile	Immobile	Immobile	Immobile	-1 dmg/ 7 days	-1 dmg/ 3 days
Dying (Dmg > HP)	Immobile	Immobile	immobile	Immobile	-1 dmg/ 7 days	-1 dmg/ 3 days

EQUIPMENT

Weapon Skills

There are adjustments on Combat Skills and Cascade skills that need to be made. On the first column are the Cepheus Engine SRD p.29 cascade skills, the second column, the Mneme Column, are its equivalent in Mneme. Mneme reorganizes the skills to have more use and value.

Traveller SRD	MNEME Parent Skill	Notes
Gun Combat: Archery	Missile Weapons	Crossbows, Bows,
Gun Combat: Shotgun	Gun Combat	
Gun Combat: Slug Pistol	Gun Combat	
Gun Combat: Slug Rifle	Gun Combat	
Melee Combat: Bludgeoning	Melee Combat	
Melee Combat: Natural	Melee Combat	Includes Unarmed Combat
Melee Combat: Piercing	Melee Combat	
Melee Combat: Slashing	Melee Combat	
	Melee Combat	Shields
	Special	Nets, Whips, Blowguns
Athletics	Athletics	Throwing grenades or objects

Alternate Cascade Skills

An alternative breakdown of Cascade Skills, swapping Gun Combat into two types: Rifle and Pistol.

MNEME	Cascade	Notes
Melee Combat	Impact	Any hafted weapons wielded 1 or 2-handed.
Melee Combat	Polearms	Any pole mounted weapon, staves, or Spears.
Melee Combat	Swords	A fast rate of attack weapon
Melee Combat	Fencing	Double the defense modifier of Fencing when the character has the Advantage. Includes the Jian, Rapier, Sabers, Scimitars, and other types of Finesse weapons.
Melee Combat	Striking	Punching, Kicking, headbutting, Kneeing, Elbowing, or any unarmed Striking maneuver.
Melee Combat	Grappling	Locks, Shoots, Grabs, Twists, Pins, Wrenching, or any other techniques to manipulate the leverage and joints of an adversary.
Melee Combat	Shields	This raises the cover granted by a shield by the skill level. As well grant cover (without the bonus) to a number of adjacent allies equal to the skill.
Melee Combat	Chain	3-section staff, Sickle&Chain, chain whips (not covered)
Ranged Combat	Slings	Slings, Staff-Slings, Atl-Atls, Woomera, or Spear throwers.
Ranged Combat	Throwing	Any thrown weapon
Ranged Combat	Archery	Bows and Slingshots.
Firearm Combat	Crossbow	Gastraphetes, Chokuno, or Any Bow mounted on a platform that can be operated by one man.
Firearm Combat	Rifle	A firearm becomes a rifle when it can be effectively wielded 2-handed, like pistols with carbine platforms or modifications giving it shoulder stocks.
Firearm Combat	Pistol	One handed firearm

Equipment Trait and Properties

symbol means number.

Weapon Technology Level
The minimum Tech Level the weapon appears.

Aim Modifier
Weapons with an Aim Modifier Add this modifier to any attacks with the weapon when they perform an Aim action.

Rate of Fire or Attacks
Weapons have a Rate of Fire or Rate of Attacks property. This determines the maximum number of damage rolls it can deal per significant action.

- Weapons have a minimum number of attacks of 1 per round.
- Skill modifies the number of attacks.
  - S/2 means 1 attack, plus half of skill per round. A skill of 1 doesn't4 add any additional attacks, but a skill of 2 and 4 add respectively a second and 3rd attack.
  - S means 1 attack plus Skill.
  - number preceding the S means number of additional attacks plus Skill.
- Dual Weapon Fighting grants additional attacks opportunities. When determining the maximum amount of attacks the character is capable.
- Examples
  - A character with two daggers would have up to 3S x2 of attacks.
  - A character with an Hand Axe and Shield can have up to S + S/2 of attacks. the other attack would come from the shield.
  - A Character with a Longsword and a Short sword would have S + 3S attacks.

Slashing Damage
This is cutting, slicing, or edged weapon damage. Certain items are more vulnerable to Slashing Damage - like the gel-like property of muscles animal bodies and some fiber materials like wood or plant stalks.

Piercing Damage
Piercing damage is penetrating with a narrow point type of damage. Most living creatures and electronic components are more vulnerable to Piercing damage than mechanic items.

Bludgeoning Damage
Bludgeoning Damage is blunt force, crushing, and concussive damage that impart force into a large area relative to the target. It is more effective against hard (high Armor) but brittle targets. Its less effective against Fibrous or the gel-like properties of muscles and animal bodies.
Bludgeoning Damage can cause Knockback and Knockdown.

Armor Trait
Armors are listed in the Equipment, and they Grant Armor Rating to Characters. Also note that Attacking Weapons and Equipment have their own AR rating and Hit Points. See Armor

Minor Loading Trait (mLoading)
minor action to Load. Without performing the Load action the weapon cannot fire or be used.

Significant Loading trait (sLoading)
Significant action to load the weapon. Without performing the Load action the weapon cannot fire or be used.

Full Action Loading trait (fLoading #)
Full Action to load a weapon. Without performing the Load action the weapon cannot fire or be used.

Minor Ready trait (mReady)
A minor action to Adjust the reach of this weapon. Unready this weapon attacks at a disadvantage.

Significant Ready trait (sReady)
It takes a significant action to ready this weapon. Unready this weapon cannot be used.

Auto # Trait
Multiplies the Number of Attacks and Damage Roll per effect with attacks with Auto.

- Example
  - A 3S weapon with Autofire 2 can have up 3 success effect 2 = 3 damage rolls x Auto for a total of 6 damage rolls."

Burst # Trait
multiply the number of damage rolls per success.
Like auto, but this is for weapons with only a limited shot.

2-handed Trait
Weapon requires 2-hands, disadvantage to rolls one handed.

Versatile Trait
Weapon can be used one handed or two handed or other usage modes (example Quarterstaff). Its a free action, once per round, to change usage.

Reach # Trait
The number of Spaces in addition to the Adjacent space the weapon can attack. Most weapons can adjust their grip to from reach 1 to just adjacent.
- Disadvantage against Adversaries that are closer than its reach.
  - example a Reach 2 weapon able attack 3 spaces away suffer a disadvantage to attack adversaries 2 spaces away or adjacent.
- Versatile Weapons can change Reach as a Free Action once per round.
- Moving Attack option to adjust distance.
- mReady can be used to shift weapons reach, like weapons with a Reach 2 to just reach 1.

Range str #/# Trait
Range is determined by the strength the weapon is built for, the higher the strength the greater the range.
- The values are multiplied to Strength to get the ranges.
- Half the Damage beyond the first Range Value
- Second value is the maximum distance
- Cannot use a Weapon with str required is > Weilder's Str+2.
- Disadvantage using a weapon> Weilder's str."

Thrown #/# Trait
the two values are the multiplier to Strength in meters.
- Half the Damage beyond the first Range Value
- Second value is the maximum distance

Heavy Trait
- Disadvantage using this weapon if str is < Str 9.
- Cannot use this weapon if Str < 7."

Finesse Trait
- This weapon must be used with Dexterity to gain its benefit.
- When making Targeted Attacks in melee, reduce conditions (like cover or concealment) by 1 OR armor by 1.

Light Trait
- Can be used as an Offhand weapon. Offhand weapons grant their rate of attacks to the total Rate of Attacks.

Explosive Trait
- Explosive weapons are less effective farther away, divide the damage of explosions by distance in meters.
- at default treat Explosive Damage like Bludgeoning Damage as it is able to cause Knockback and Knockdown.

Armor Piercing Trait
Armor Piercing (AP) reduces the Effectiveness of Armor. Reduce the Armor of the Target by the AP.
- Ranged Weapons can be given Armor Piercing Qualities at a higher cost.
- Arrows/Bolts X2 Cost, AP 1
- Pistols and Submachine Guns X2 Cost AP3
- Rifles X2 Cost A

EQUIPMENT TABLE

Type	Weapon	TL	Aim	Dmg	RoA	Armor	Wt (Kg)	Cost	Notes
Melee Combat	Battleaxe	TL 0		3d3-2 sl	S/2		2.0 kg	Cr 90	Versatile (3d3 sl, RoA S, 2-handed), 1 AR, 11 hp
Melee Combat	Club	TL 0		2d3-1 bl	2S		1.0 kg	Cr 3	Light, 1 AR, 8 hp
Melee Combat	Dagger	TL 0		2d3-1 pi	3S		0.5 kg	Cr 45	Finesse, light, thrown 1/3, 1 AR, 7 hp
Melee Combat	Greatclub	TL 0		3d3-2 bl	S/2		5.0 kg	Cr 6	2-handed, 1 AR, 14 hp
Melee Combat	Handaxe	TL 0		1d6 sl	S		1.0 kg	Cr 60	Light, thrown 1/3, 7 AR, 8 hp
Melee Combat	Javelin	TL 0		1d6 pi	S/2		1.0 kg	Cr 15	Thrown str 1.5/7, 1 AR, 8 hp
Melee Combat	Mace	TL 0		1d6 bl	S		2.0 kg	Cr 75	Heavy, 7 AR, 11 hp
Melee Combat	Maul	TL 0		4d3 bl	S/2		5.0 kg	Cr 150	Heavy, 2-handed, 1 AR, 14 hp
Melee Combat	Quarterstaff	TL 0		1d6 bl	2S		2.0 kg	Cr 6	Versatile, 2-handed, Note: Default is being held in the middle (3d3-2 bl, RoA S, 2-handed, Reach 1, note: being held like a 2-handed spear), 2 AR, 11 hp
Melee Combat	Spear	TL 0		1d6 pi	S/2		1.5 kg	Cr 60	Thrown 1/3, Versatile (3d3-2 pi, RoA S, 2-handed, Reach 1), 1 AR, 10 hp
Melee Combat	Unarmed Strikes	TL 0		1d2	S				Versatility: Kicks (RoA S/2, 1d3 bl)
Melee Combat	Warhammer	TL 0		3d3-2 bl	S/2		1.0 kg	Cr 135	Versatile (3d3 bl, RoA S, 2-handed), 1 AR, 8 hp
Melee Combat	Whip	TL 0		2d3-1 sl	S/2		1.5 kg	Cr 45	Finesse, Reach 1, 1 AR, 10 hp

Type	Weapon	TL	Aim	Dmg	RoA	Armor	Wt (Kg)	Cost	Notes
Melee Combat	Flail	TL 1		3d3-2 bl	S/2		1.0 kg	Cr 150	Flail, 1 AR, 8 hp
Melee Combat	Glaive	TL 1		3d3 sl	S/2		4.0 kg	Cr 300	reach 1, 2-handed, 1 AR, 13 hp
Melee Combat	Greataxe	TL 1		2d6 sl	S/2		3.5 kg	Cr 350	2-handed, Reach 1, 1 AR, 13 hp
Melee Combat	Halberd	TL 1		2d6-1 sl	S/2		5.0 kg	Cr 400	Heavy, Reach 1, 2-handed, 7 AR, 14 hp
Melee Combat	Lance	TL 1		3d3-1 pi	S/2		3.0 kg	Cr 250	Reach 1, Versatile (3d3 pi, RoA S, 2-handed, Reach 1), 7 AR, 12 hp
Melee Combat	Light hammer	TL 1		2d3-1 bl	S		1.0 kg	Cr 50	Light, thrown 1/3, 1 AR, 8 hp
Melee Combat	Morningstar	TL 1		3d3-2 pi	1		2.0 kg	Cr 300	Heavy, 1 AR, 11 hp
Melee Combat	Pike	TL 1		3d3 pi	S/2		9.0 kg	Cr 125	Heavy, Reach 3/2, 2-handed, 7 AR, 17 hp
Melee Combat	Shortsword	TL 1		1d6 pi	3S		0.5 kg	Cr 250	Light, 7 AR, 7 hp
Melee Combat	Sickle	TL 1		2d3-1 sl	2S		1.0 kg	Cr 100	Light, 1 AR, 8 hp
Melee Combat	Trident	TL 1		1d6 pi	S/2		2.0 kg	Cr 150	thrown 1/3, Versatile (3d3-2 pi, RoA S/2, 2-handed, Reach 1), 1 AR, 11 hp
Melee Combat	War pick	TL 1		3d3-2 pi	S		1.0 kg	Cr 125	AP 1, 1 AR, 8 hp
Melee Combat	Broadsword	TL 2		3d3-2 sl	2S		1.3 kg	Cr 350	1 AR, 9 hp
Melee Combat	Greatsword	TL 2		4d3 sl	S/2		3.0 kg	Cr 600	Heavy, Reach 2/1, 2-handed, 7 AR, 12 hp
Melee Combat	Longsword	TL 2		3d3-2 sl	S		1.5 kg	Cr 500	Versatile (3d3 sl, RoA 2S, Reach 1, 2-handed), 7 AR, 10 hp
Melee Combat	Rapier	TL 2		3d3-2 pi	3S		1.0 kg	Cr 350	Finesse, 1 AR, 8 hp
Melee Combat	Scimitar	TL 2		1d6 sl	3S		1.5 kg	Cr 350	Finesse, 7 AR, 10 hp

Type	Missile Weapons	TL	Aim	Dmg	RoA	Armor	Wt (Kg)	Cost	Notes
Missile	Sling	TL 0	Aim+0	2d3-1 bl	1		0.1 kg	Cr 3	mLoading, range str 1.5/6, 2 AR, 4 hp
Athletics	Dart	TL 1	Aim+0	2d3-1 pi	S		0.5 kg	Cr 8	Thrown 1/3
Athletics	Net	TL 0	Aim+0	NA	1		1.5 kg	Cr 15	Special, Thrown str 0.3/0.7
Missile	Crossbow, light	TL 2	Aim+2	3d3-2 pi	1		2.5 kg	Cr 60	sLoading, 2-handed, Bolt 1 Cr 0.02kg, 1 AR, 11 hp
Missile	Shortbow	TL 0	Aim+1	1d6 pi	1		1.0 kg	Cr 90	2-handed range str 4/15 , Arrow 1 Cr 0.05kg, 1 AR, 8 hp
Missile	Crossbow	TL 2	Aim+2	3d3 pi	1		9.0 kg	Cr 150	fLoading 3, 2-handed, range str 5/20, Bolt 1 Cr 0.02kg, 1 AR, 17 hp
Missile	Composite Bow	TL 0	Aim+1	3d3-2 pi	1		1.5 kg	Cr 500	2-handed, range str 7/30, Arrow 1 Cr 0.05kg, 1 AR, 10 hp
Missile	Longbow	TL 0	Aim+1	3d3-2 pi	1		1.0 kg	Cr 500	2-handed, range str 7/30, Arrow 1 Cr 0.05kg, 1 AR, 8 hp
Missile	Crossbow, hand	TL 2	Aim+1	1d6 pi	1		1.5 kg	Cr 150	sLoading, range str 1.5/6, Small Bolt 0.5 Cr 0.02kg, 1 AR, 10 hp

Type	Firearms	TL	Aim	Dmg	RoA	Armor	Wt (Kg)	Cost	Notes
Gun Combat	Auto Pistol 7x25mm	TL 6	Aim+0	1d6 pi	2S		1.00 kg	Cr 100	(200m, /2000m), 10 AR, 8 hp
Ammo	Auto Pistol Mag 7x25mm 10 rnds	TL 6					0.10 kg	Cr 3	mLoading
Gun Combat	Auto Pistol 12x20mm	TL 6	Aim+0	2d3 pi	2S		1.50 kg	Cr 100	(150m, /1500m), 10 AR, 10 hp
Ammo	Auto Pistol Mag 12x20mm 7 rnds	TL 6					0.25 kg	Cr 3	mLoading
Gun Combat	Compact Auto Pistol 10x20mm	TL 7	Aim+0	1d6+1 pi	2S		1.00 kg	Cr 100	(150m, /1500m), 10 AR, 8 hp
Ammo	Auto Pistol Mag 10x20mm 8 rnds	TL 7					0.20 kg	Cr 3	mLoading
Gun Combat	Auto Pistol 10x20mm	TL 7	Aim+0	1d6+1 pi	2S		1.50 kg	Cr 100	(150m, /1500m), 10 AR, 10 hp
Ammo	Auto Pistol Mag 10x20mm 15 rnds	TL 7					0.25 kg	Cr 3	mLoading
Gun Combat	Submachine Gun 12x20mm	TL 6	Aim+1	3d3 pi	3S		7.50 kg	Cr 230	Auto 2, (150m, /1500m), 10 AR, 16 hp
Ammo	Submachine Gun Mag 12x20mm 20 rnds	TL 6					1.00 kg	Cr 10	mLoading
Ammo	Submachine Gun Mag 12mm 50 rnds	TL 6					2.50 kg	Cr 25	mLoading
Gun Combat	Shotgun 20x50mm	TL 6	Aim+1	1d3 pi	S		4.00 kg	Cr 200	Burst 3, (25m, /400m), 1 AR, 13 hp
Ammo	Shotgun 20x50mm cartridges	TL 6					0.02 kg	Cr 1	mLoading
Gun Combat	Automatic Rifle 7×60mm	TL 6	Aim+3	4d6-3 pi	3S		4.50 kg	Cr 250	(1000m, /5000m), 10 AR, 14 hp
Ammo	Clip 7x60mm 8 rnds	TL 6					0.25 kg	Cr 12	mLoading
Gun Combat	Assault Rifle 5x50mm	TL 7	Aim+3	3d6-2 pi	3S		3.00 kg	Cr 600	Auto 3, (500m, /3000m), 10 AR, 12 hp
Ammo	Mag 5x50mm 30 rnds	TL 7					0.50 kg	Cr 30	mLoading
Gun Combat	Assault Rifle 7x40mm	TL 7	Aim+2	3d6 pi	3S		3.50 kg	Cr 400	Auto 3, (500m, /3000m), 10 AR, 13 hp
Ammo	Mag 7x40mm 30 rnds	TL 7					0.70 kg	Cr 35	mLoading
Ammo	Mag 7x40mm AP5 30 rnds	TL 7					0.70 kg	Cr 70	
Gun Combat	Light Machine Gun 7x60mm	TL 7	Aim+3	4d6-3 pi	3S		15.0 kg	Cr 4,000	Auto 4, (1000m, /4000m), 10 AR, 21 hp
Ammo	Light Machine Gun 7x60mm mag 50 rnds	TL 7					2.5 kg	Cr 70	sLoading

Type	Explosives	TL	Aim	Dmg	RoA	Armor	Wt (Kg)	Cost	Notes
Athletics	Stick Grenade	TL 6		5d3 ex	S		0.50 kg	Cr 20	mReady,Thrown 1/3
Athletics	Hand Grenade	TL 7		7d3 ex	S		0.25 kg	Cr 50	mReady,Thrown 1.5/5

Type	TL 6 to TL 8 Armor	TL	Aim	Dmg	RoA	Armor	Wt (Kg)	Cost	Notes
Armor	Light Plate	TL 6				3 AR	4.00 kg	Cr 3,000	Steel Plates covering the torso and head
Armor	Heavy Plate	TL 6				7 AR	10.00 kg	Cr 7,000	Heavy steel plates covering the torso and head.
Armor	Frag Vest	TL 7				3 AR	4.00 kg	Cr 3,000	Steel plates and plastic woven into synthetic fiber. Head and Torso only.
Armor	Ballistic Fabric	TL 7				2 AR	4.00 kg	Cr 2,000	flexible but heavy synthetic fiber armor. Head and Torso only.
Armor	Ballistic Plates	TL 7				11 AR	11.50 kg	Cr 11,000	synthetic fiber holding specially made plates. Head and Torso only.
Armor	Ballistic Plates	TL 8				12 AR	12.50 kg	Cr 12,000	synthetic fiber holding specially made plates. Head and Torso only.
Armor	Assault Armor	TL 8				15 AR	8.00 kg	Cr 15,000	synthetic fiber holding specially made ceramic and steel plates. Head and Torso only.

Type	Shield	TL	Aim	Dmg	RoA	Armor	Wt (Kg)		Notes
Melee Combat	Buckler	TL 1		1d3 bl	S/2		1.0 kg	Cr 30	Grants Cover 1 if skilled., 1 AR, 8 hp
Melee Combat	Small Shield	TL 1		1d3 bl	S/2		2.0 kg	Cr 60	Grants Cover 1 if skilled., 1 AR, 11 hp
Melee Combat	Medium Shield	TL 1		2d3-1 bl	S/2		4.0 kg	Cr 120	Cover 1, 1 AR, 13 hp
Melee Combat	Large Shield	TL 1		2d3 bl	S/2		8.0 kg	Cr 240	Cover 2, 1 AR, 17 hp

Type	TL0 to TL3 Armor	TL	Aim	Dmg	RoA	Armor	Wt (Kg)	Cost	Notes
Armor	Leather	TL 0				1 AR	7.0 kg	Cr 50	Hardened leather full body.
Armor	Cloth	TL 0				1 AR	7.0 kg	Cr 50	Thick layers of cloth full body.
Armor	Scale	TL 1				2 AR	35.0 kg	Cr 1,500	Lamellars or Overlapping Scale full body.
Armor	Mail	TL 2				2 AR	20.0 kg	Cr 2,500	Only 1 AR against Bl damage full body
Armor	Brigandine	TL 3				3 AR	25.0 kg	Cr 5,000	Overlapping plates under leather full body.

Type	Equipment	TL	Aim	Dmg	RoA	Armor	Wt (Kg)	Cost	Notes
Gear	Backpack, 60L Leather	TL 0					4.0 kg	Cr 250	30kg capacity, rain proof
Gear	Backpack, 40L Leather	TL 0					3.0 kg	Cr 180	20kg capacity, rain proof
Gear	Backpack, 100L Canvas	TL 1					3.9 kg	Cr 233	50kg capacity, Rain proof at TL6
Gear	Backpack, 60L Canvas	TL 1					2.8 kg	Cr 166	30kg capacity, Rain proof at TL6
Gear	Backpack, 30L Canvas	TL 1					1.7 kg	Cr 104	15kg capacity, Rain proof at TL6
Gear	Load Bearing Gear (Webbing, Belts, Straps)	TL 0					0.5 kg	Cr 10	2L worth of pouches, loops, and belts and bandoliers to hold it.
Gear	Waterskin 5L	TL 0					5.7 kg	Cr 45	flexible container to 5L of wine or water.
Gear	Waterskin 1L	TL 0					1.2 kg	Cr 10	Small container for water.
Gear	Wooden Cask 15L	TL 2					5.5 kg	Cr 80	Holds 15L of Liquid
Gear	Bed Roll and Blanket, Light	TL 0					4.0 kg	Cr 50	made up of Furs in earlier TL, made of more durable and comfy materials in higher TLs.
Gear	Metal Pot (5L)	TL 1					3.0 kg	Cr 200	Used to Boil and Make food. Typically made of Copper in earlier eras, later made of Iron.
Gear	Earthen Pot (15L)	TL 0					10.0 kg	Cr 10	Used to Boil and Make food. Fragile but easy to replace.

Type	Equipment	TL	Aim	Dmg	RoA	Armor	Wt (Kg)	Cost	Notes
Gear	Outdoor Clothing/Travel Clothes	TL 0					2.0 kg	Cr 50	Typically rugged and durable material. From furs, heavy linen tunics, to modern hiking clothes. Includes headgear, sandals, shoes or boots, and some small pouches.
Gear	Water Purification	TL 7					0.5 kg	Cr 100	Can process up to 100L of water.
Gear	Survival Basics	TL 0					0.0 kg	Cr 2	from waterproof Leather pouches to keep flammable tinder, flint and steel, lighters, small cup or bowl and eating utensils. Sometimes a place to hold salt, herbs found, and spices.
Gear	Hooded Lamps	TL 0					0.5 kg	Cr 10	24 hours per 1L of lamp oil.
Gear	Bullseye Lanterns	TL 1					1.0 kg	Cr 20	24 hours per 1L of lamp oil.
Gear	Flashlight 5m L4	TL 6					0.3 kg	Cr 10	Number of hours x TL-5. $1 batteries
Gear	Flashlight 7m L4	TL 6					0.5 kg	Cr 20	Number of hours x TL-5. $2 batteries
Gear	Earthen Bottle 1L	TL 0					2.0 kg	Cr 2	wrapped with cords and wicker.
Gear	Wooden/Bamboo Bottle 1L	TL 1					1.2 kg	Cr 2	corded and wide mouth
Gear	Light Intensifier Goggles	TL 7						Cr 500	Reduce darkness by 4
Gear	Infrared Goggles	TL 6						Cr 500	Reduce darkness by 2 if there is heat signature.
Gear	Personal Communicator 60km	TL 8					0.3 kg	Cr 250	
Gear	Hand Radio 1.5 km	TL 6					2.0 kg	Cr 1,000	
Gear	Fiber Rope 10mm	TL 0					1.0 kg	Cr 3	10m units, 120kg capacity.
Gear	Fiber Rope 20mm	TL 0					2.0 kg	Cr 30	10m units, 500kg capacity

Type	Equipment	TL	Aim	Dmg	RoA	Armor	Wt (Kg)	Cost	Notes
Gear	Day's Ration	Any					0.5 kg	Cr 1	a days worth of meal for an active person of mostly starch, fat, and dried fruit.
Gear	Dried Meats/Fish	TL 1					0.5 kg	Cr 5	Dried proteins - either in sausages, jerky, smoked, or meats ground into pemmican.
Gear	Medical Consumables	Any					0.5 kg	Cr 100	See Medical Consumables
Gear	Trauma Kits TL3, 10 uses	TL 3					1.5 kg	Cr 300	DM+3 to First Aid
Gear	Trauma Kit TL3-2, 10 uses	TL 1					1.0 kg	Cr 2,000	DM+3 to First Aid
Gear	Trauma Kit TL6-1, 10 uses	TL 5					2.5 kg	Cr 5,000	DM+6 to First Aid
Gear	Trauma Kit TL7, 1 use	TL 7					0.4 kg	Cr 70	DM+7 to First Aid

Non Combat Equipment

Advanced / Cheap Technology

Advanced and Cheap Technology Options are meant to reflect goods, equipment, or services that overlap between Technology Levels. Advanced and Highly Advanced technologies are available but expensive and with some quirks. Older Technology are still available but cheaper.

- Requires Referees Approval.
- Note some Cultures, People, and Places have different TL for different Industries and Areas. In the Developing world a cellphone can be found in the boondocks but transportation, medicine, food, are all far below TL8.

Advanced Technology

Take Goods, Services, or Equipment available in a higher TL that would appear in its earliest form in the previous TL. It is Available in the Previous TL at 10x the cost.

- A TL item is now available in TL-1.
  - Notation is the Items Original TL with -1.
  - Example a TL9 computer available in TL8 is Computer TL9-1.
- Cost is 10x
- There will be a drawback typically its
  - Fragile 1/2 HP or AR or Both.
  - It has Limited Uses. Example 1/10 the work life.
  - It has a Skill level requirement
  - Its Difficulty to Repair is 2 more to repair.
  - Bulkier x3 Weight and Volume.

Highly Advanced Technology

This is Advanced Technology taken to a greater level, allowing technology 2 levels greater. Examples would be stone age surgery or electroplating.

- A TL item is now available in TL-2.
  - Notation is the Items Original TL with -2.
  - Example a TL9 computer available in TL7 is Computer TL9-2
- Cost is 100x
- There will be a drawback is more extreme
  - Fragile 1/4 HP or AR or Both.
  - It has a Very Limited Use. Example 1/100 the work life.
  - It has a Skill level requirement
  - Its Difficulty to Repair is 4 more.
  - Bulkier x10 Weight and Volume.

Cheap Technology (Cheap Tech)

Take Goods, Services, or Equipment available in a lower TL and have more Advanced Manufacturing and Mature Markets commoditize these - making them very cheap OR gain a specific benefit. They typically have a special quirk or side benefit from being made in a more advanced market.

- A TL item is now available cheaply in TL+1.
  - Notation is the Items Original TL with +1.
  - Example a TL8 computer manufactured in TL9 is Computer TL8+1.
- Choose an Example Benefits:
  - Cost is 1/10
  - Hardened - vs EMP, AR/HP increased, various Proofing for Water, Radiation, Vacuum, etc..
  - Extended Work Life, x3.
  - Lighter 1/3 Weight and Size.

Very Cheap Technology (Very Cheap Tech)

- A TL item is now available cheaply in TL+2.
  - Notation is the Items Original TL with +2.
  - Example a TL7 computer manufactured in TL9 is Computer TL7+2.
- Choose an Example Benefits:
  - Cost is 1/100
  - Hardened against 3 different conditions - vs EMP, AR/HP increased, various Proofing for Water, Radiation, Vacuum, etc..
  - Extended Work Life, x10.
  - Lighter 1/10 Weight and Size.

Medical Consumables TL

To simplify Bookkeeping allocate a budget on medical consumables any one kind or purpose does not make up more than 20% of the Total of Consumables.

- Assume per 100Cr there is 0.5KG of medicine (weight being from the Packaging).
  - All Prices assume TL7.
  - **Antibiotics** - available in TL6 in the form of penicillin and in earlier eras as tribal knowledge. TL8 10Cr per week of treatment.
  - **Antiseptics** - weaken and slow the growth of bacteria, this is a very common type of medicine even in TL0. TL8 1Cr per day of applications.
  - **Analgesics** - medicine for pain relief to buy time for the body to heal. TL8 1Cr per day of applications.

- **Antivirals** - a medicine used to treat viral infections. Available as early as TL6, but it wasn't very until much later TLs and against only certain virus strains.
- **Vaccines** - Vaccines appear as early as TL4. Some vaccines serve as a booster for what is mostly inoculated or is partially effective in a related or similar disease.
- **Wound Dressing** - available as early as TL0 with any fabric or leaves used in keeping wounds from getting infected, stem bleeding, and manage pain.
- **Tools and Applicators**. injections, pads, 1Cr per day.
- **Antihistamines**. Manage allergic symptoms when the body's response does more harm than good.
- **Anti Inflammatory drugs**. These manage inflammation when the body's response does more harm than good.

- **Medical Consumables and their Care**.
  - Light Care. Light use of Analgesics, Wound Treatment, and Antiseptics. 1cr a day.
  - Moderate Care. more powerful Analgesics, extensive wound treatment, Antiseptics, Antibiotics. 2cr a day
  - Critical Care. 5cr per day.
  - Intensive Care. 10cr per day.
- Step 1. Set a budget how much Medicines will be purchased.
- Step 2. Indicate an amount of Medical consumables Credits equal to the budget and Corresponding weight.
  - Example. 250Cr is 2.5x 100Cr, which makes for 1.24kg of Medicine.
- Step 3. When there is Wounds
- Example.
  - The character needs Wound Dressing and treatment-related Medical Consumables. The Party's Medical Consumables is worth 200Cr, up to 40Cr worth of Wound Treatment and Dressing,

Stimulants

Stimulants allow the character to ignore a number of wound penalties from Exhaustion Damage equal to the level. This is more powerful than several doses of Coffee or Caffeine and also more dangerous.

- Example
  - Stims 6 hours Level 1. 25cr 2000pc/kg
  - Stims 6 hours Level 2. 75cr 2000pc/kg
- Side Effect.
  - Take note of the Levels of Stimulants. When the character exceeds their Endurance DM +1 of Stimulants taken per 24 hours they take 1 exhaustion damage.
  - Loading up in stimulants deals damage but stimulants ignore damage, while the character ignores the Penalties of the stimulants once it wears off all that pain and damage will be felt.
- This is an Addictive and Dependency causing type of drug. Continuous use alters behavior and "withdrawal" can cause DM penalties or Disadvantage.

Pain Medication

Pain medication either block Pain or induce euphoria that blocks pain.

- Example: powerful analgesics and legal opioids.
  - Pain Medication 6 hours Level 1. 10cr
  - Pain Medication 6 hours Level 2. 30cr
- Take note of the Levels of Pain medication. When the character exceeds their Endurance DM +1 of Pain medication taken per 24 hours they take 1 exhaustion damage.
- Loading up in Pain medication deals damage but stimulants ignore damage, while the character ignores the Penalties of the stimulants once it wears off all that pain and damage will be felt.
- This is an Addictive and Dependency causing type of drug. Continuous use alters behavior and "withdrawal" can cause DM penalties or Disadvantage.

Trauma Kits

- These are specialized First Aid kits to treat Serious, Critical, and Intensive Injury.
- The Trauma Kit's TL determines its Dice Modifier to the Medic's Roll for First Aid.
- A Trauma Kit's Cost is equal to the Modifier it Grants times Cr 10 times the number of uses.
  - A Use is the duration the Kit is being applied to the Target. So in the 30 rounds to apply First Aid, the Trauma Kit's DM modifier applies to all those roll.s
  - A Trauma kit can have a DM of TL+1 at x10 its cost like in Advanced Technology.
  - A Trauma kit can have a TL+2 at x100 its Cost like in Highly Advanced Technology.
  - Using the Medical Consumables Rules for weight.

- Examples:
  - TL3 Trauma Kit with 10 uses. 300 Cr 1.5kg.
  - TL1 Trauma Kit+2 with 10 use. 2,000 Cr 1kg.
  - TL5 Trauma Kit+1 with 10 uses. 5000 Cr 2.5kg
  - TL7 Trauma Kit with 1 use. 70 Cr.

Cover Material Table

This assumes only the Area of the Attack from a cannon or firearm will destroy to get at its Target behind it.

Cover	AR	HP
Light Wood (Pine) Door 30cm High x 30cm Wide x 10cm Thick	0	9
Light Wood (Pine)Wall 30cm High x 30cm Wide x 20cm Thick	0	14
Medium Wood Door (Mahogany) 30cm High x 30cm Wide x 10cm Thick	1	13
Medium Wood Wall (Mahogany) 30cm High x 30cm Wide x 10cm Thick	1	19
Heavy Wood Door (Oak) 30cm High x 30cm Wide x 20cm Thick	1	16
Heavy Wood Wall (Oak) 30cm High x 30cm Wide x 20cm Thick	1	24
Brick Wall 30cm High x 30cm Wide x 20cm Thick	3	25
Concrete Wall 30cm High x 30cm Wide x 20cm Thick	4	28
Iron Door or Hull 30cm High x 30cm Wide x 0.2cm Thick (2mm)	8	9
Iron Door or Hull 30cm High x 30cm Wide x 2cm Thick (20mm)	8	19
Steel Door or Hull 30cm High x 30cm Wide x 0.2cm Thick (20mm)	10	19
Aluminum Door or Hull 30cm High x 30cm Wide x 0.5cm Thick (5mm)	3	8
Aluminum or Hull 30cm High x 30cm Wide x 0.2cm Thick (20mm)	3	13

QUICK CHARACTERS

To quickly Generate Characters without Rolling using Character Points.

Instead of rolling to determine if the character Qualifies, Survives, becomes a Commissioned Officer, Advances, or suffers Aging Penalties The character just pays an amount of character points (cp). Typically equal to the Difficulty of Roll. *Example a End 6+ roll costs 6cp*.

Difference with Rolling Characters

- Cannot select Jack-of-All Trades Skill and Same Skill or Stat more than once per term.
- Maximum Skill level is 1/2 the number of terms rounded up and cannot have a skill greater than 4.
  - Example. a 5 Term character can only have a Skill level of 2 at the highest.
- Benefits have a CP cost.

Step 1. Determining the amount of Points for the Game. 75 for a low level game or 100 cp for more powerful characters. *Example a 75cp game.*

Points	Power Level	Age	Age
75 cp	2 Terms with Commission and Advancement. 4 Term with no advancement and poor stats.	26	34
100 cp	4 Terms with Commission and Advancement, 6 Terms with no advancement and surviving aging and poor stats.	34	42
125 cp	6 Terms with Commission and Advancement, 8 Terms with no advancement and surviving aging and poor stats.	42	50
150 cp	8 Terms with Commissions and Advancements. 10 terms of no advancement and poor stats.	50	58

Step 2. Assign CP to Stats.

STATS	RAISE by 1	DM	Raise By	Total Cost
5	0 cp	-1	0	0
6	1 cp	0	+1	1 cp
7	2 cp	0	+2	3 cp
8	3 cp	0	+3	6 cp
9	4 cp	+1	+4	10 cp
10	5 cp	+1	+5	15 cp
11	6 cp	+1	+6	21 cp
12	7 cp	+2	+7	28 cp

Raise the Stat by 1

Example A character named Ren has a Str 6 (1cp), Dex 9 (6cp), End 6 (1cp), Int 6 (1cp), Edu 5 (0cp), Soc 5 (0cp) for 9cp, 66cp left.

Step 3. Determine Dice Modifier.

Example: Ren's stats are Str 6 (DM 0), Dex 9 (DM +1), End 6 (DM 0), Int 6 (DM 0), Edu 5 (DM -1), Soc 5 (DM -1).

Step 4. Determine Background Skills. Get EDU DM + 3.

Example: Edu 5 (DM-1) +3 = 2 Background Skills.

Step 5. Determine Home World. see CESRD p. 26 or see the *Quick Character Creation Diagram*.

Example. Industrial World- Broker and Electronics.

Step 6. Determine Career and Spend CP equal its Qualification.

- Every change Career adds 2 CP to the cost to qualify.
- Drifter has no CP Cost to Qualify.

Example. Ren spends 5cp to Qualify for Rogue. 61cp left.

Step 7. Gains Basic Training.

Streetwise 0, Mechanics 0, Gun Combat 0, Melee Combat 0, Recon 0, and Vehicle 0.

Step 8. Pay CP to survive, character gains 1 skill or 2 (if Athlete, Barbarian, Belter, Drifter, Entertainer, Hunter, or Scout).

Example: Ren pays 4cp to Survive. Ren gains Melee Combat 1. 57cp remaining.

Step 9. Commissioned Officer. If the Career has Commission, the Character can choose to spend the Cp equal to the difficulty to become an Commissioned officer. Ignore this step when done.

- This is optional and characters don't need to spend on Commission.

Example. Ren pays 6cp to be a Commissioned Officer. 51cp remaining.

Step 10. Advancement. Careers with Commission allow for advancement in ranks. Advancement grants a skill and increase is rank. The cost is CP equal to the Advancement Difficulty.

- This is optional and characters don't need to spend CP for advancement.

Example. Ren pays 7cp to advance as an Associate. Ren gains a second skill, streetwise 1, in their first term 44cp remaining.

Step 11. Check for Aging. Is the character over 34 years old or more than 4 terms.

- Suffer -1s to Mental and Physical Stats of your choice equal to Terms-4 per Term past 4.
- You can pay CP 1cp per stat to prevent the reduction.

Example Ren is 18, but will end 22 years old after this term.

Step 12. Decide to Re-enlist, Muster out, or Change Career.

Step 12.A. Change Career. Character spends CP to Qualify. They proceed to Basic Training in Step 7.

Step 12.B. Re-Enlist. Character spends CP the Re-enlist difficulty to re-enlist. They proceed to step 8 noting the term.

Step 12.C. Muster Out. Mustering out

- Character gains an amount of Benefits equal to the Terms they Survive.
  - +1 more Benefit at Rank >3. Example. A character with Rank 6 has 3 additional benefits.
- Character has to Spend CP equal to x2 the roll to gain the benefit.
  - Example, if a benefit requires a roll of 4 then it costs 8cp.
- Cannot get a benefit more than once.
- Any Remaining Character Points can be Contacts.

Example. Ren decides to Re-enlist. he spends 4cp to re-enlist. 40cp remaining. Ren begins at step 8.

***Step 8 2nd Term**. Ren pays 4cp to survive and gains Recon 1. 34cp remaining. Ren skips Step 9.*

***Step 10 2nd Term**. Ren pays 7cp to Advance, and gains the Rank Soldier and Gun Combat 1. Ren chooses the Dex +1 for a total of 10. 27cp remaining.* Ren ignores step 11.

***Step 12 2nd Term**. Ren decides to Re-enlist spending 4cp 23cp remaining.*

***Step 8 3rd Term**. Ren pays 4cp to survive and gains Computer 0. 19cp remaining.* Ren skips Step 9.

***Step 10 3rd Term**. Ren pays 7cp to Advance, and gains the Rank Lieutenant. Ren chooses the Dex +1 for a total of 11. 12cp remaining.* Ren ignores step 11.

Step 12 3rd Term. Ren decides to Muster Out with 12cp remaining. Ren spends 2 cp for 1,000, 4cp 5,000 CR. Ren spends 6cp Various Contacts.

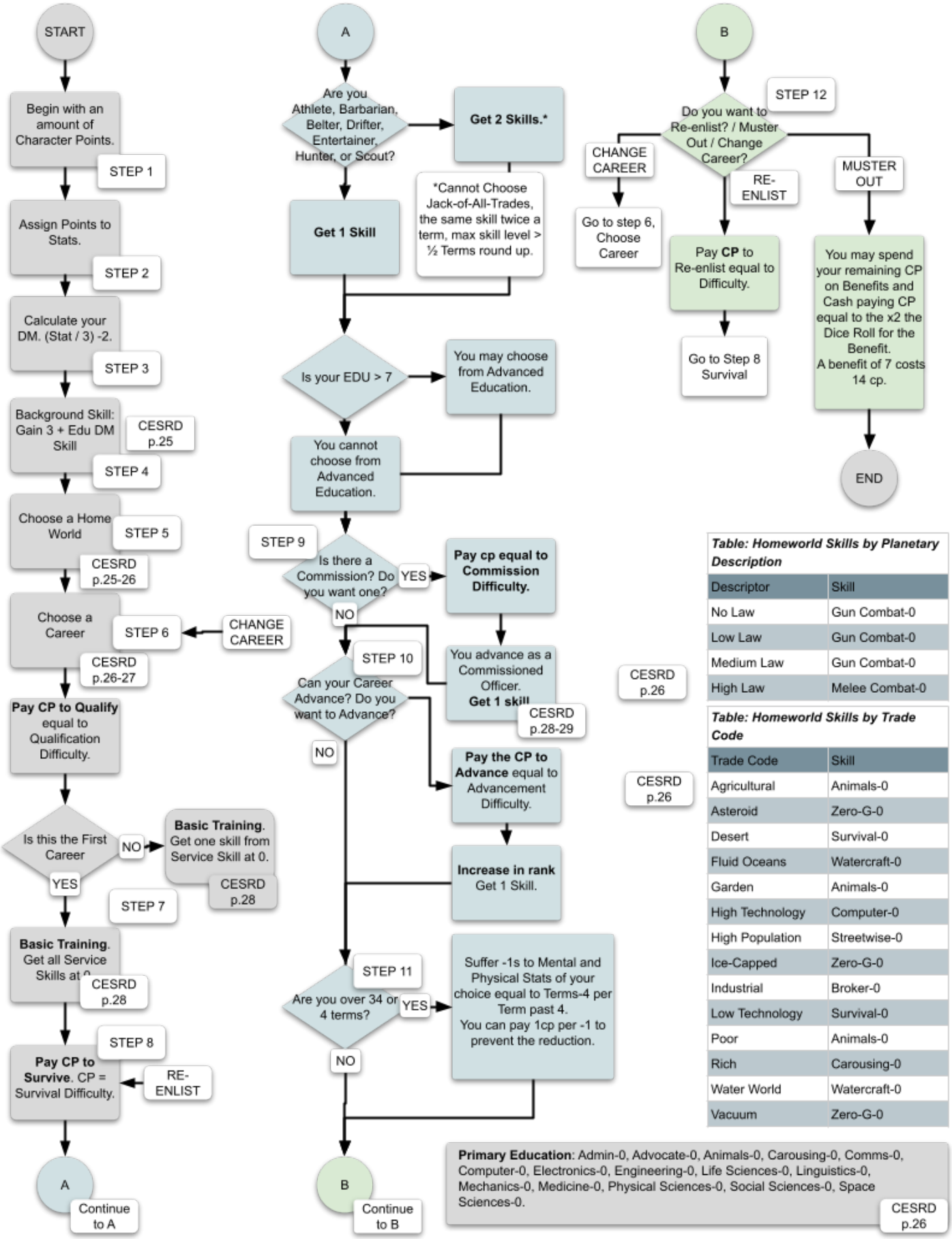


Table: Homeworld Skills by Planetary Description

Descriptor	Skill
No Law	Gun Combat-0
Low Law	Gun Combat-0
Medium Law	Gun Combat-0
High Law	Melee Combat-0

CESRD p.26

Table: Homeworld Skills by Trade Code

Trade Code	Skill
Agricultural	Animals-0
Asteroid	Zero-G-0
Desert	Survival-0
Fluid Oceans	Watercraft-0
Garden	Animals-0
High Technology	Computer-0
High Population	Streetwise-0
Ice-Capped	Zero-G-0
Industrial	Broker-0
Low Technology	Survival-0
Poor	Animals-0
Rich	Carousing-0
Water World	Watercraft-0
Vacuum	Zero-G-0

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Primary Education: Admin-0, Advocate-0, Animals-0, Carousing-0, Comms-0, Computer-0, Electronics-0, Engineering-0, Life Sciences-0, Linguistics-0, Mechanics-0, Medicine-0, Physical Sciences-0, Social Sciences-0, Space Sciences-0.

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Sample Encounter

A simple encounter where a larger and fast force is trying to pursue a smaller but better equipped force.

Two Hands of Yeomen Warriors, in their war glory have been summoned by their Liege Lord, Za Ergan.*
Outworlders have been spotted in the lord's domain, and are escaping on foot in the wilderness. Their lord's voice - Asvar Enede has called for his own Yeomen but they are going to circle around the wilderness to cut off the outworlders. Enede warns them of Outworld high artifice, and that their quarry's senses may be more acute than a skan-lion.*
***
Underboss Valch was not happy with the betrayal of that upstart Ergan and his agent turd-for-brains Enede. These rat-hole barbarians should have stuck to the deal and handed over the prisoner for the goods. To change their mind after all that work and without even bargaining. Valch was lucky his mules were the only that died, the bluff worked but these primitives arent as dumb as the Cooperative hoped. Now they have to escape dragging the asset back.
*A title

Game Preparation.

- Key Considerations
  - Keep note of the Movement. The Yeomen have 4/10/15 1.5m spaces move at light and the Enforcers may have 3/7/10 1.5 space move because of medium load.
- Yeomen should outnumber the Enforcers 3 to 1. With reinforcements coming if they fight in the open.
- Navigating while running is a combined difficulty of Survival, Navigation, or Recon and Athletics.
- **Narrative Combat.**
  - Wilderness have obstacles: bogs, quick-sand, covered broken ground that reduce move by 1-2 spaces per round.
  - Yeomen can try to enter Wilderness at an angle and try to intercept.
- **Tactical Combat.**
  - 4.5m Scale per space (hex or square) would be best. this is 3x the 1.5m scale per space, or 1/3 movement in spaces. On a 25 x 25 space map where each space is 4.5m the Enforcers with 1/2/3 space move will be overtaken by the Yeomen with
    - Enforcers have 2 Round Lead. They would have moved 6 spaces (27m) into the map by the time the Yeoman begin. They can be exchanging fire at this round.
    - On Round 5, when yeomen have a run of 5 spaces per round, would have caught up. This is 15 spaces into a 25 x 25 space map.

Objectives

- Enforcers to Escape and Reach the edge of the map or break the morale of the Yeomen if they took more than half casualties. This assumes the prisoner is alive.
- Yeomen to recapture the prisoner. Forcing the Enforcers to abandon the prisoner by either encircling them with superior numbers or breaking their morale by taking some of them down.

Description for Playing Yeomen

Earlier in the day Asvar Enade explains that the Outworlders are in the domain and that they should be ready to mount their bisons to ride them down. Many Yeomen have been summoned, they come ever hour.
When the call to arms did come, Wildrunners have called their crows with them and their wistles tell the Yeomen the Outworlders are fleeing to the east. A hundred breaths away from the back of a bison, towards the Wilderness bordering the mountains.
Asvar Tells the Yeomen to capture or kill them. 5 slaves to the yeoman who catches one alive, and 2 for each one dead. The runaway Bondsman returned alive is 20 slaves, no reward if dead.

Description Playing Enforcers

The Escape Baloon was hidden among the drop pod compartments in the mountains, it should reach a kilometer elevation near explosively in a couple of minutes. This would allow the ship to pick them out without having to land.
The job is simple: get to the escape baloon before the natives know any better with the package.

Conditions and other Factors.

- Yeoman have bisons mounts that allow them to skirt the perimeter of the Wilderness. Wilderness is to dense and broken for bison mounts to be used.

- Yeomen are light Encumbrance.
- Bisons
  - Wear AR 2, HP 10/20/30, Size +2 (as big as a draft horse, wearing a wooden saddle that spreads the weight on rump and shoulder).
  - move 3 spaces (4.5m), dash 7 spaces (10m), and Run at 10 spaces (15m), much like a Medium Load Human.
- There are 10 Yeoman have to recapture the prisoner.
- The prisoner is a wild card.
- The Under-Enforcers need to reach their emergency extraction point - an emergency balloon that will carry them out of range of the Za Ergan and make it easy for pickup. .
- There are 3-4 Enforcers each with up to 6,000cr budget for equipment and gear. Note encumbrance
- Typical scenario is that the Enforcers have enter the wilderness with a small lead.

Starting the Game

- Depending on the size of the map or the budget of time, have the Yeomen arrive 2 turns after the Enforcers.
- If Enforcers just ambush near the edge of the wilderness and wilderness for cover enforcers will circle around.

Yeoman - Warrior

Yeoman - Warrior

Str	9	+1	**Int**	6	0	**HP** 0.3	8
Dex	6	0	**Edu**	5	-1	**HP** 0.67	16
End	9	+1	**Soc**	5	-1	**HP**	24

TL6-3 Barbarian (3 terms) 30 years old
Animals (Farming) 0, Gun-Combat-0 (Archery) Mechanics 0, Gun Combat (Slug Rifle) 1, Melee Combat, Recon 1, Survival 0, Animals (Riding) 0, Athletics 0, Tactics 0.
Assault Rifle 7x40mm Aim+2 3d6 pi 3S [(500m, /3000m), 10 AR, 13 hp] 1 pc, Mag 7x40mm 30 rnds [mLoading] 5 pc,
Auto Pistol 7x25mm Aim+0 1d6 pi 2S [(200m, /2000m), 10 AR, 8 hp] 1 pc, Auto Pistol Mag 7x25mm 10 rnds [mLoading] 3 pc,
Cheap Shortsword 1d6 pi 3S [Light, 7 AR, 7 hp] 1 pc,

A TL6 Yeoman
Yeoman Warrior LOADOUT: Load Bearing Gear (Webbing, Belts, Straps) [2L worth of pouches, loops, and belts and bandoliers to hold it.] 1 pc, Flashlight 5m L4 [Number of hours x TL-5. $1 batteries] 1 pc, Wooden/Bamboo Bottle 1L [corded and wide mouth] 1 pc, Outdoor Clothing/Travel Clothes [Typically rugged and durable material. From furs, heavy linen tunics, to modern hiking clothes. Includes headgear, sandals, shoes or boots, and some small pouches.] 1 pc, Qty, 11.8 kg, 379.8 Cr, 620.3 Cr

Step	Term #	Event	Points
Step 1	Term 0	50cp Character	50 cp
Step 2	Term 0	Str 8 (3cp), Dex 6 (1cp), End 8 (3cp), Int 6 (1cp), Edu 5 (0cp), Soc (0cp) for 7cp,	-9 cp
Step 3	Term 0	Str 8 (DM 0), Dex 6 (DM 0p), End 8 (DM 0), Int 6 (DM 0), Edu 5 (DM -1), Soc (DM -1) for 7cp,	
Step 4	Term 0	2 Background Skills	
Step 5	Term 0	Low Law Poor World: Animals-0, Gun Combat 0	
Step 6	Term 1	Barbarian 5cp	-5 cp
Step 7	Term 1	Basic Training. Mechanics, Gun Combat, Melee Combat, Recon, Survival, Animals	
Step 8	Term 1	Survives - 2 skills: Gun Combat 1, Recon 1	-6 cp
Step 9	Term 1	Commission NA	
Step 10	Term 1	Advancement NA	
Step 11	Term 1	Aging NA	
Step 12	Term 2	Re-Enlist	-5 cp
Step 8	Term 2	Survives - 2 skills: Strength, Endurance	-6 cp
Step 9	Term 2	Commission NA	
Step 10	Term 2	Advancement NA	
Step 11	Term 2	Aging NA	
Step 12	Term 3	Re-Enlist	-5 cp
Step 8	Term 3	Survives - 2 skills: Athletics, Tactics	-6 cp
Step 9	Term 3	Commission NA	
Step 10	Term 3	Advancement NA	
Step 11	Term 3	Aging NA	
Step 12		Muster Out. 3 benefits. Weapon 4cp, Cash 1,000 Cr 4cp	-8 cp
		REMAINING CP	**0 cp**

Under-Enforcer - Specialist

Under-Enforcer

Str	8	-0	**Int**	7	0	**HP** 0.3	7
Dex	9	+1	**Edu**	5	-1	**HP** 0.67	14
End	6	0	**Soc**	5	-1	**HP**	23

TL9-2 Rogue (3 terms) 30 years old
Broker-0, Advocate-0
Streetwise 1, Mechanics 0, Gun Combat 0, Melee Combat 0, Recon 0, Vehicle 0, Electronics 0.
Dagger 2d3-1 pi 3S [Finesse, light, thrown 1/3, 1 AR, 7 hp] 1 pc,
Auto Pistol 12x20mm Aim+0 2d3 pi 2S [(150m, /1500m), 10 AR, 10 hp] 1 pc, Auto Pistol Mag 12x20mm 7 rnds [mLoading] 3 pc,
Assault Rifle 5x50mm Aim+3 3d6-2 pi 3S [(500m, /3000m), 10 AR, 12 hp] 1 pc, Mag 5x50mm 30 rnds [mLoading] 7 pc,
Hand Grenade 7d3 ex S [mReady,Thrown 1.5/5] 2 pc,
Cheap Ballistic Plates 12 AR [synthetic fiber holding specially made plates. Head and Torso only.] 1 pc

Under-Enforcer LOADOUT: Flashlight 5m L4 [Number of hours x TL-5. $1 batteries] 1 pc, Waterskin 1L [Small container for water.] 1 pc, Outdoor Clothing/Travel Clothes [Typically rugged and durable material. From furs, heavy linen tunics, to modern hiking clothes. Includes headgear, sandals, shoes or boots, and some small pouches.] 1 pc, Light Intensifier Goggles [Reduce darkness by 4] 1 pc, Personal Communicator [] 1 pc, Qty, 24.7 kg, 3,082.5 Cr, 2,917.5 Cr

Step	Term #	Event	Points
Step 1	Term 0	50cp Character	50 cp
Step 2	Term 0	Str 8 (3cp), Dex 8 (3cp), End 6 (1cp), Int 6 (1cp), Edu 5 (0cp), Soc (0cp) for 7cp,	-9 cp
Step 3	Term 0	Str 8 (DM 0), Dex 8 (DM 0), End 6 (DM 0), Int 6 (DM 0), Edu 5 (DM -1), Soc (DM -1)	
Step 4	Term 0	2 Background Skills	
Step 5	Term 0	High Population - industrial - Broker-0, Advocate-0	
Step 6	Term 1	Rogue 5cp	-5 cp
Step 7	Term 1	Basic Training. Streetwise, Mechanics, Gun Combat, Melee Combat, Recon, Vehicle	
Step 8	Term 1	Survives : Streetwise-1	-6 cp
Step 9	Term 1	Commission NA	
Step 10	Term 1	Advancement NA	
Step 11	Term 1	Aging NA	
Step 12	Term 2	Re-Enlist	-4 cp
Step 8	Term 2	Survives - Electronics 0	-6 cp
Step 9	Term 2	Commission NA	
Step 10	Term 2	Advancement NA	
Step 11	Term 2	Aging NA	
Step 12	Term 3	Re-Enlist	-4 cp
Step 8	Term 3	Survives - Dex +1	-6 cp
Step 9	Term 3	Commission NA	
Step 10	Term 3	Advancement NA	
Step 11	Term 3	Aging NA	
Step 12		Muster Out. 3 benefits. 4cp for 5,000Cr; 2cp for 2,000Cr; 4cp Int +1	10 cp
		REMAINING CP	**20 cp**

Designers Notes and Referee's Guide

Key Principles

Control of Forces.

Control means knowing how to concentrate, divide, reserve, support, direct, and apply forces.

Concentration or Division of Force.

The more Action and Attention one force has, the more they can adapt to their Adversary. Concentrating and Dividing Force is about shifting ones strength (a concentration of firepower) and one's weakness (their flanks and blind areas) to discover and exploit the strengths and weaknesses of the other party.

The most common Tactic is breaking forces into 3s - the reserve or Rear guard, the main force and the vanguard. The purpose of which is to test and adapt to the actions and pace of the adversary and conditions.
Example:
If one force encounters another opposing force, their secondary force will try to outflank or move outside the line of sight of the opposing force.

"To be strong everywhere is to be weak everywhere"
Sun Tzu

Initiative and Control

Prepared and rehearsed Actions taken can shift the initiative and control to one party from another. There

is reacting to the opposing force: to defend, hunker down, to return fire, to displace, to divide etc… and there is a surplus of attention when this is done so well and executed so efficiently that one can employ actions that cause the other party to abort their actions and plans.

Reacting and Taking control comes before Prediction of Behavior, living to survive and fight is better than learning theory only to be human and wrong and die.

"Know yourSelf" - Sun Tzu

Predicting Behavior and Outcomes

When one has had practice Prediction becomes the next principle to learn. What is not explained and often overlooked is that Prediction is not about getting it right, but being less wrong.
Its begins with Predictions where the risks of the action taken are not so leathal. Devastating or Decisive outcomes are great for stories and heroes, but as people develop the practice predicting is about calculated risks.

"Know your enemy" - Sun Tzu
"The more you know, the more you realize you don't know" - Aristotle

Does the Referee have to be a Great Tactician?

"Perfect is the Enemy of Good"
The Referee can

- add or reduce adversaries in the middle of the game,
- he can rule in favor or against the players,
- he can drop modifiers or conditions
- This is because stranger things have been known to happen even in real world combat.

They don't have to be a great tactician - but they do need to have the capacity to learn and talk to the players for mutual areas for improvement.

Other Narrative Techniques

Techniques to keep in mind, write it down in the referee notes, practice, create examples and test to see how to use it.

Use Conditions and Action to Guide Narrative

Use the Threat, Awareness, Visibility, Range Concealment, etc… to help narrate what's happening. Is a Simple formula of Condition + Verb/Action.

- In the Distance (Condition: range) Gunfire (Condition: threat) and nature violently escaping (verb).
- The Guard moves (action) closer (condition: range) to inspect (action).
- The enforcer stands their full height (Action), displaying their hands close to their weapons (Action). (Implies a threat).

Players Reiteration

Let other Players reiterate the conditions. This is an Active listening technique.

- Buys the Referee time to think and get their voice back.
- Allows the Referee to correct the understanding of other Players.

Slow down and Simplify

We can simplify a scene to speed up the action, but we can use simplification and slowing down to reduce confusion. Use conditions (Threat, Awareness, Visibility, Range, Concealment, etc…) to break down a scene to slow it down for players.

- Example:
  - "The guard rushes to attack you" is simple and quick.
  - "you see the guard quickly advancing towards you." "you see them with weapons drawn, ready to use on you."
- Usage - Slowing things down:
  - Helps people used to a lot of Rolls transition to Mneme mechanics by breaking up the action for them to respond.

Tactical Combat

Using Maps, Figurines, Tokens, and Virtual Table Top Tools running tactical combat is letting the map, visibility, positioning, and pictures do the storytelling for the Referee and players so everyone can relieve some mental load.

Action/Attention Economy

Forces and Resources measure people and organizations, but the outcome that these try to

measure is actions and attention. Every Action requires attention to implement or perform well, while Attention without action or resources is helpless.

In TRPGs the concept of Action Economy was developed when Parties with the most Actions, not just any action but action that can impact lives and stakes, determine odds of success or failure. Some characters with more actions are more effective than other characters with less Actions.

Mneme System addresses this with the *Superiority Mechanic* and the *One Roll per turn* so that it is still about the concentration and use of force and resources while not taking too much spotlight and time for the players and referee.

Dilemma of Practice vs Theory

Guides like these only serve to give a medium for discussion what happens in the practice of any skill or discipline. Practice, running the Mneme Combat scenarios, will always be the best way to learn.

- Example:
  - *When is it BEST to divide your forces?* When the adversary concentrates their force - leaving an area of vulnerability, where you can blunt their concentrated force with delay tactics and lead them to expose a vulnerability.
  - *When is this False/Wrong* - when you cannot coordinate - and dividing further diminishes one's resources that neither force can exploit a vulnerability or escape/delay a concentrated force.
  - *When is the Above Wrong* - when the adversary will behave believing you have sufficient force to delay their concentrated force and exploit their vulnerability. That the adversary options are more advantageous than your current options.
  - *When is the Above wrong* - when the adversary's other options are more advantageous than their current options.
  - This can go on and on.

Modifiers and Conditions Affect

- **Adverse Conditions**
  - The more equally adverse conditions are for both parties - the more extreme dice results are needed to hurt each other.
- Work out the Details After,
  - Range typically is the biggest and most changing factor.
  - Next is Cover and Concealment. Time, Weather and Terrain can add 1-4 to conditions.
  - Then Size and Movement. Either parties are moving noisy and fast, or parties are moving with a low profile and slowly adding 1-2 to conditions.
- It becomes more lethal and decisive when Conditions are Asymmetric: the party with less Condition modifiers will be able to harm the other party.
  - If adversaries are Camouflaged and concealed they have raise the difficulty for the PCs.
  - If adversaries has Condition negating properties: Infrared, night vision, or sensors.
- **Keep note of Difference of Conditions**.
  - PC's what are the Conditions affecting their Difficulty. Can the PC's make those rolls.
  - Adversary what the Conditions affecting their Difficulty. If its Higher than the PCs then it impacts the PC's Fail Effect.
  - Get the Die Result of the PCs - Difficulty. Then check by subtracting the Conditions Adversaries have in harming the PCs.

How Range Scale is Calculated

Every 2 Steps is Three to the Power increased by 1 rounded up. Every Step is Times 1.5 to the power.

Range/ Size	Range Mod	NOTES
0.5 m	Melee/-3	
1.0 m	Melee/-3	
1.5 m	Melee/-3	
3 m	-2	
5 m	-1	
10 m	0	
15 m	1	
30 m	2	
50 m	3	
100 m	4	
150 m	5	
300 m	6	
500 m	7	
1.0 km	8	Limit of Most Firearms
1.50 km	9	
3 km	10	
5 km	11	
10 km	12	
15 km	13	
30 km	14	
50 km	15	
100 km	16	
150 km	17	
300 km	18	
500 km	19	
1,000 km	20	Low Earth Orbit
1,500 km	21	
3,000 km	22	Moon's Diameter
5,000 km	23	
10,000 km	24	Moon's Circumference
15,000 km	25	About the Diameter of earth
30,000 km	26	
50,000 km	27	About the Earth's circumference
100,000 km	28	
150,000 km	29	
1 Csec	30	1 light second
1.5 Csec	31	
3 Csec	32	
5 Csec	33	
10 Csec	34	
15 Csec	35	
30 Csec	36	
1 Cmin	37	

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